

Quintile Return Book

Average monthly return of each factor's quintile portfolios and the long / short quintiles selected from them. 225 factors, ordered by in-sample L–S CAGR (last 48 months, net of cost).

HOW TO READ

■ Long
 ■ Short
 ■ Neutral

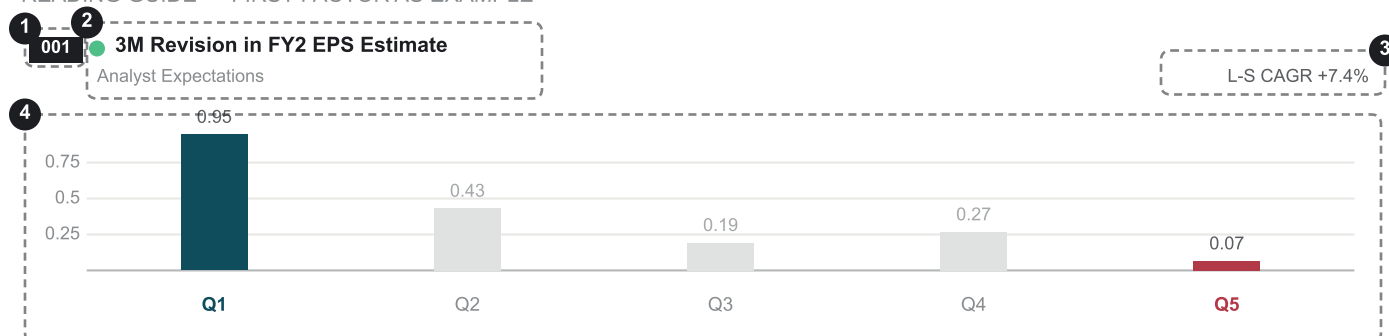
Long / short quintiles follow the live 48-month rules — spread-threshold based, not fixed to Q1 / Q5. Grey quintiles are left out of the portfolio.

FACTOR STYLES

Valuation	35	Price Momentum	50
Earnings Quality	46	Size	4
Analyst Expectations	18	Historical Growth	38
Capital Efficiency	27	Volatility	7

001 In current model portfolio (28 factors)

READING GUIDE — FIRST FACTOR AS EXAMPLE

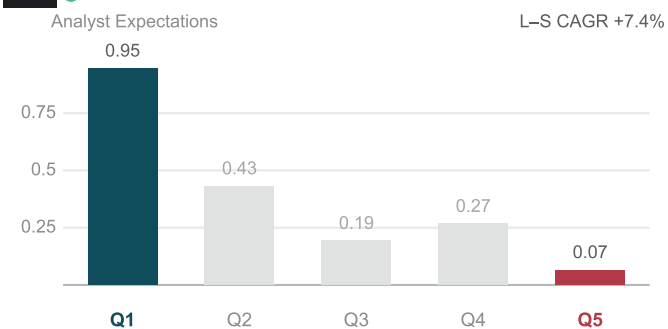
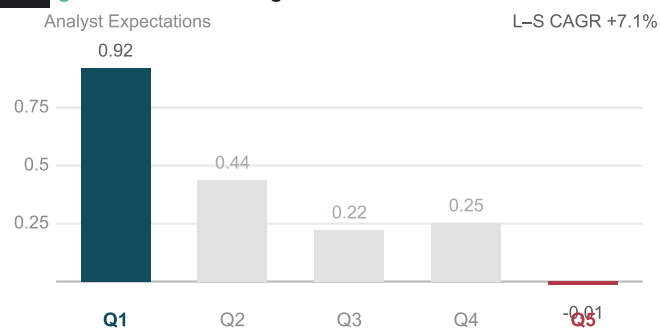
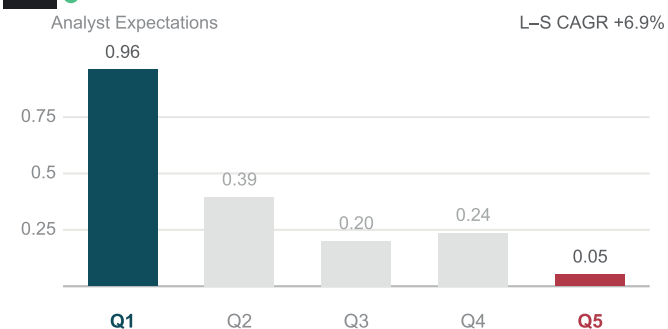
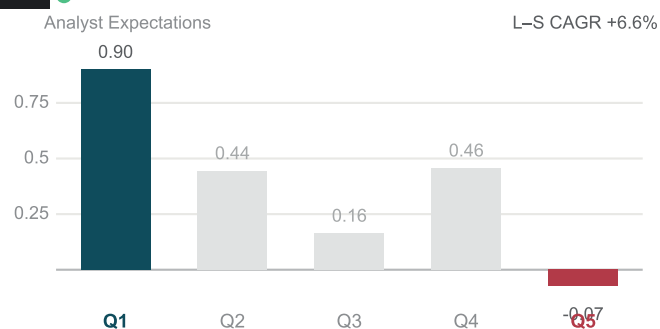
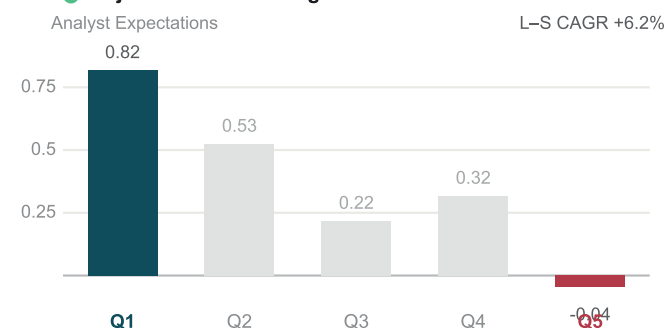
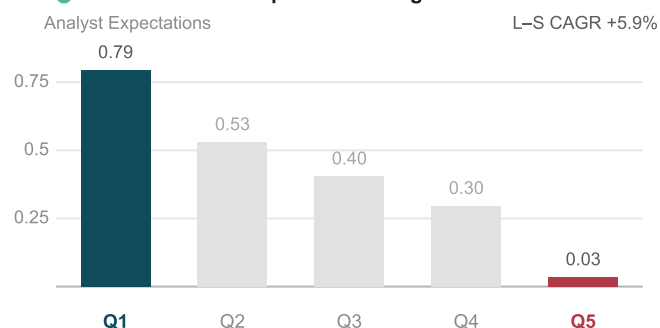


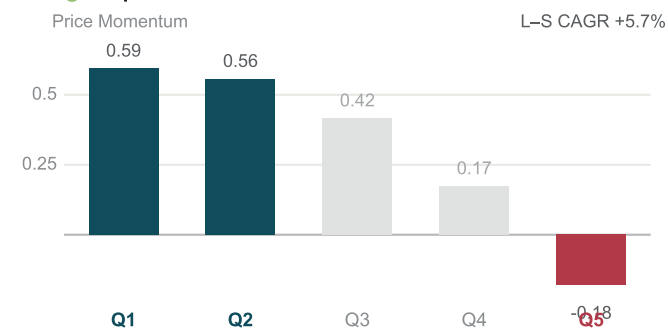
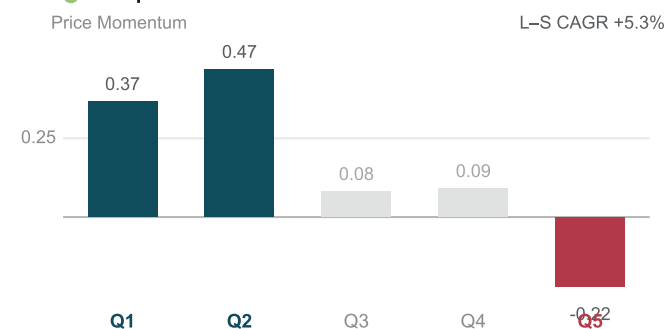
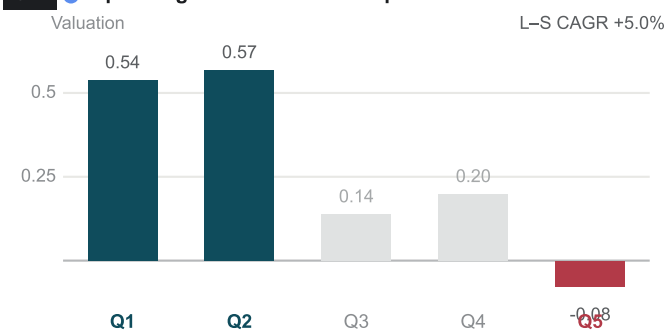
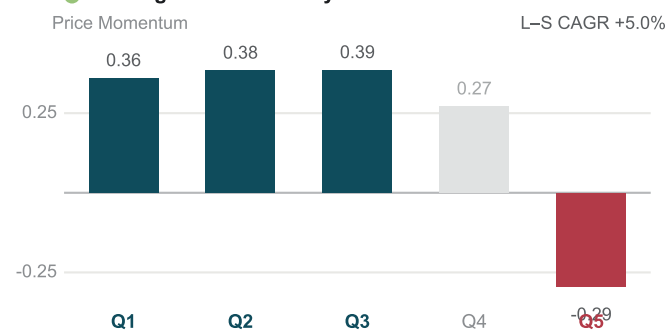
1 Rank - ordered by **3** L-S CAGR, descending.
Filled dark badge = in the current Top-50 portfolio.

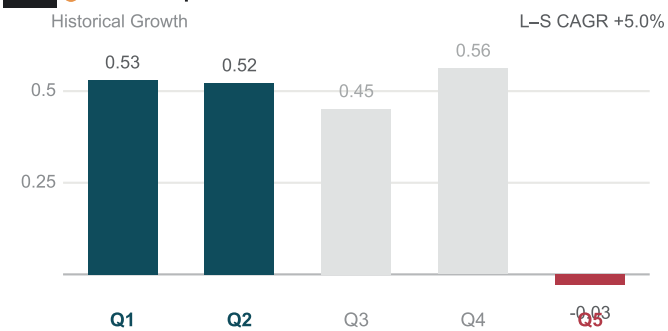
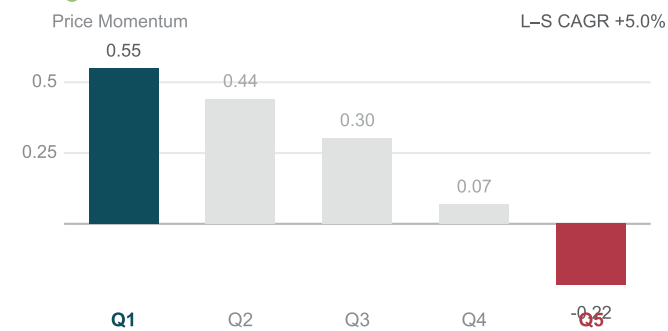
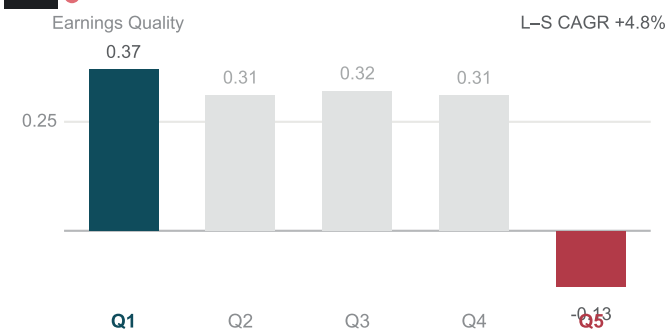
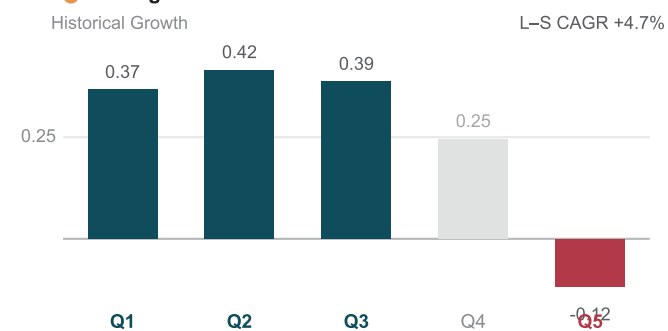
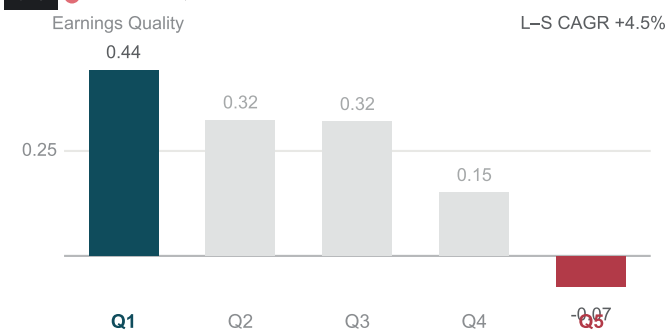
2 Factor name - colour dot = style (legend above).

3 L-S CAGR - LAST 48 MONTHS only (rolling IS), net of 20 bp cost. The live selection / sort basis. The L/S labels and sector exclusions shown are these same live 48M rules.

4 Bars - FULL HISTORY (Jul 2015 - Jul 2026): avg monthly return of the five quintiles, BEFORE costs. L/S tags are the live 48M rules; only the return window differs from **3** (so signs can disagree).

001 ● 3M Revision in FY2 EPS Estimate**002 ● Street Revision Magnitude****003 ● 3M Revision in FY1 EPS Estimate****004 ● Number of EPS FY1 Revisions****005 ● Adjusted Revision Magnitude****006 ● Standardized Unexpected Earnings**

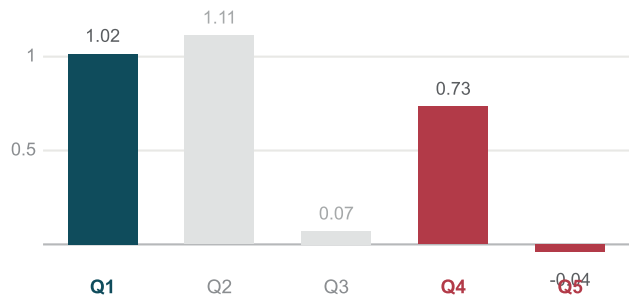
007 ● **6M Chg in Target Price**008 ● **Slope of 52W Price Trend**009 ● **Sharpe Ratio**010 ● **Cash Flow Return on Invested Capital**011 ● **Operating Cash Flow to Enterprise Value**012 ● **Closing Price to 260 Day Low**

013 Dividend per Share Growth**014 24M Residual Return Variance****015 3Y Chg in Earnings per Share****016 ROE 20 Qtr Standard Deviation****017 1Y Chg in Sales Turnover****018 ROA 20 Qtr Standard Deviation**

019 ● **Adjusted Number of EPS FY1 Revisions**

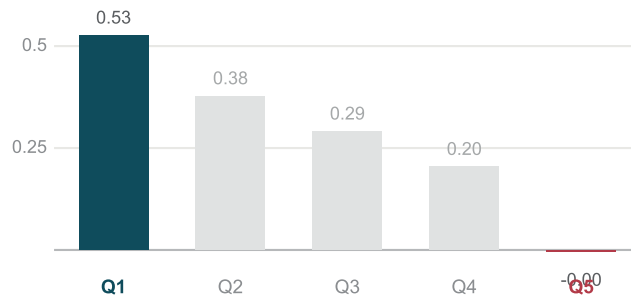
Analyst Expectations

L-S CAGR +4.5%

020 ● **Operating Cash Flow to Equity**

Capital Efficiency

L-S CAGR +4.4%

021 ● **Free Cash Flow to Sales**

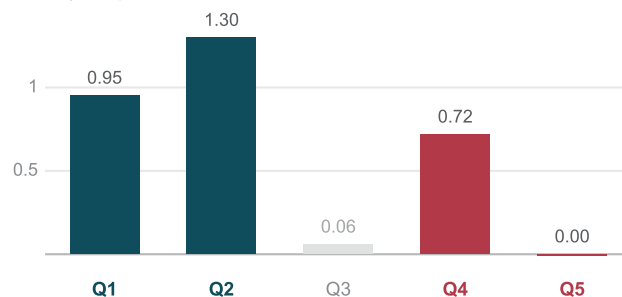
Earnings Quality

L-S CAGR +4.3%

022 ● **Analyst Earnings Estimate Diffusion**

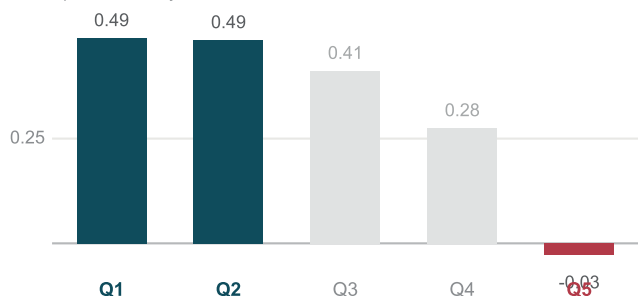
Analyst Expectations

L-S CAGR +4.3%

023 ● **Operating Cash Flow Ratio**

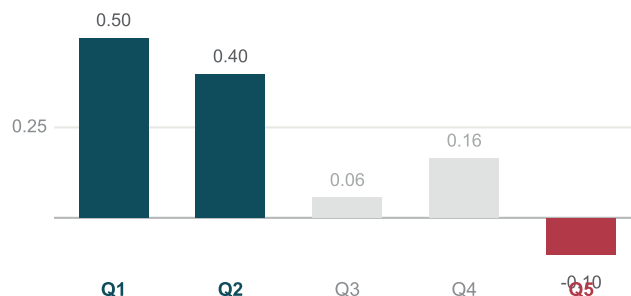
Capital Efficiency

L-S CAGR +4.3%

024 ● **Sales to Price Ratio**

Valuation

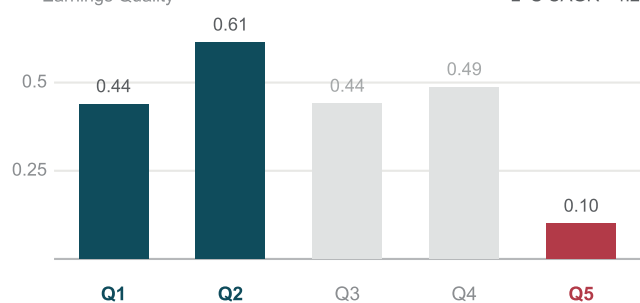
L-S CAGR +4.3%



025 ● Net Profit Margin

Earnings Quality

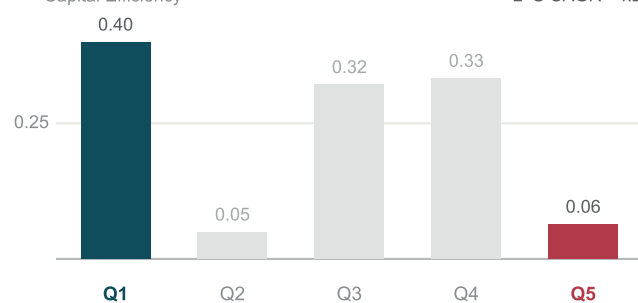
L-S CAGR +4.2%



026 ● Long Term Debt to Equity Ratio

Capital Efficiency

L-S CAGR +4.2%



027 ● 1Y Chg in Operating Cash Flow per Share

Historical Growth

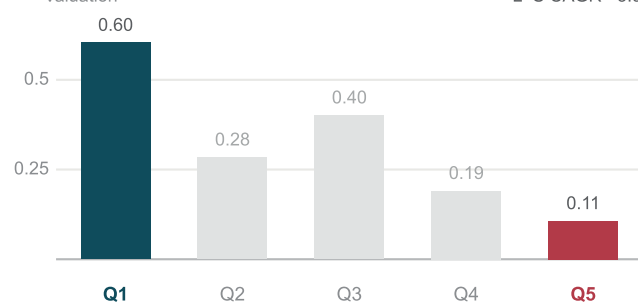
L-S CAGR +4.2%



028 ● Free Cash Flow to Common Equity

Valuation

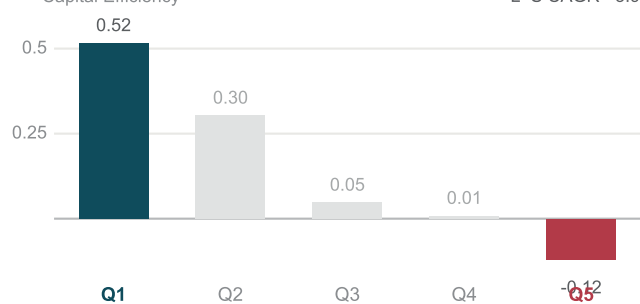
L-S CAGR +3.9%



029 ● Tobin's Q Ratio

Capital Efficiency

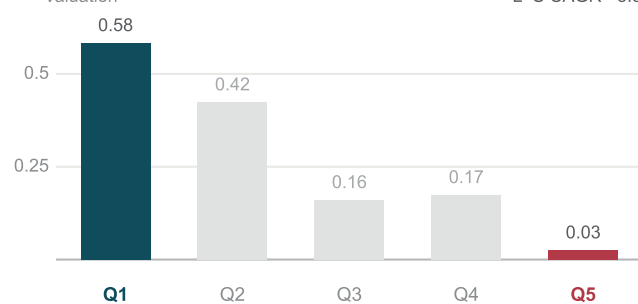
L-S CAGR +3.9%

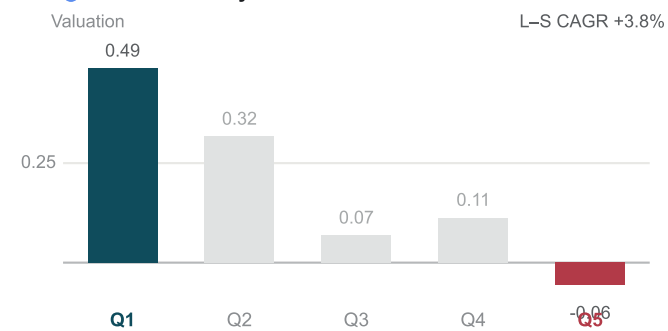
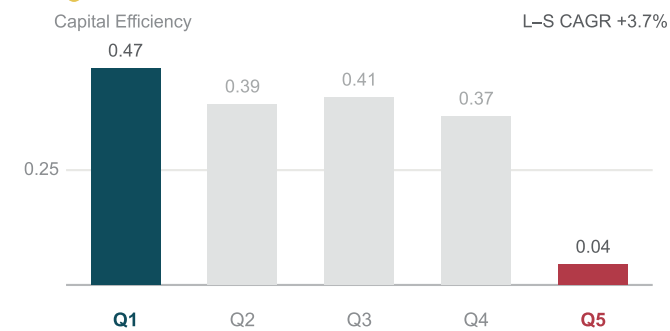
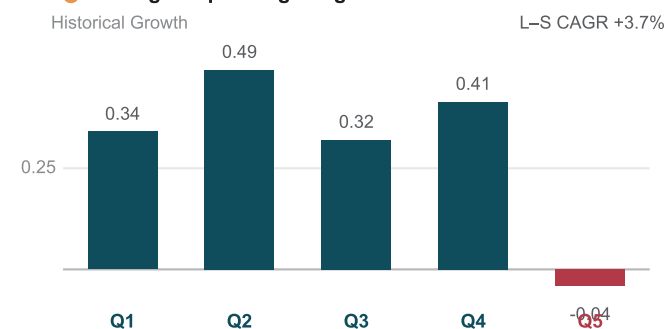
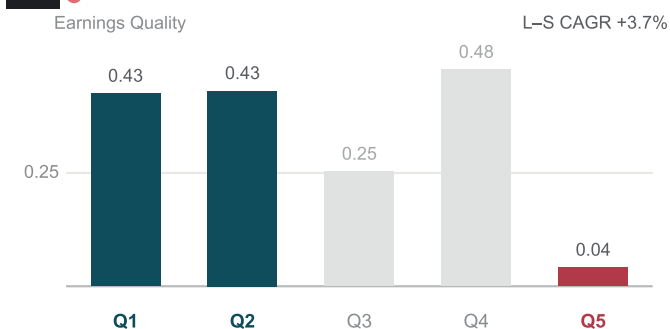
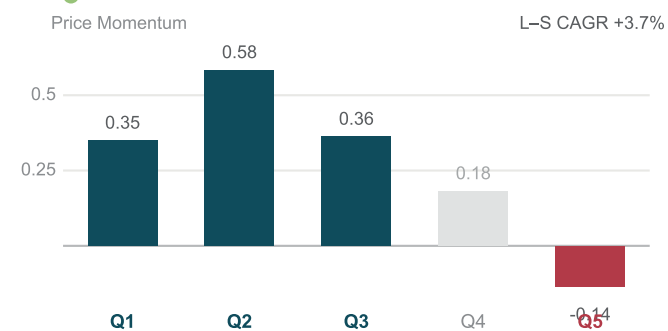
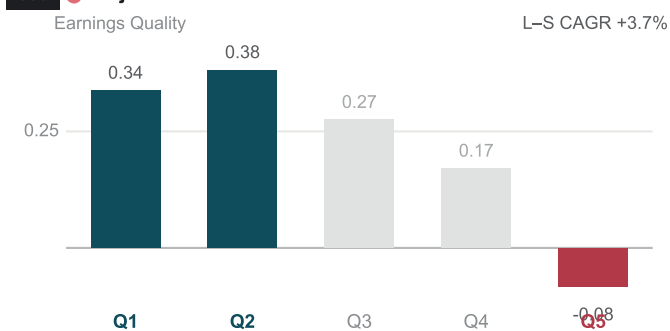


030 ● Operating Cash Flow to Price

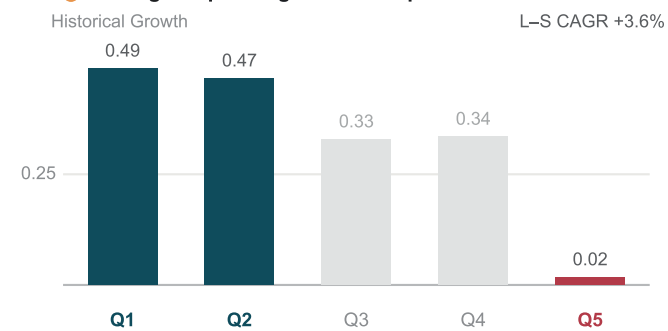
Valuation

L-S CAGR +3.8%

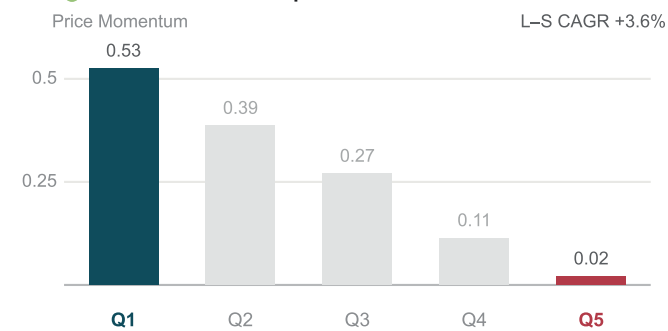


031 ● **Current Liability to Price**032 ● **Gross Profit to Assets**033 ● **3Y Chg in Operating Margin**034 ● **Asset Turnover**035 ● **180D Price TStat**036 ● **Adjusted Accruals**

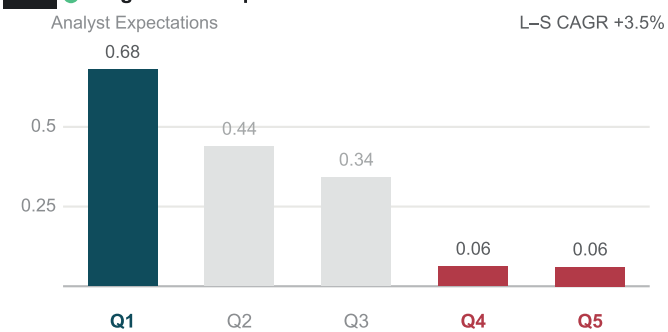
037 ● 3Y Chg in Operating Cash Flow per Share



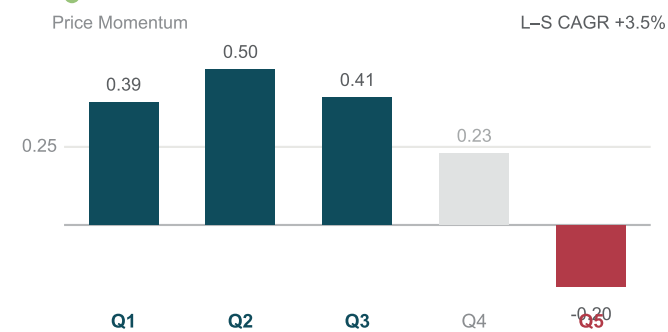
038 ● Institution Ownership Concentration



039 ● Target Price Gap to 6M EMA



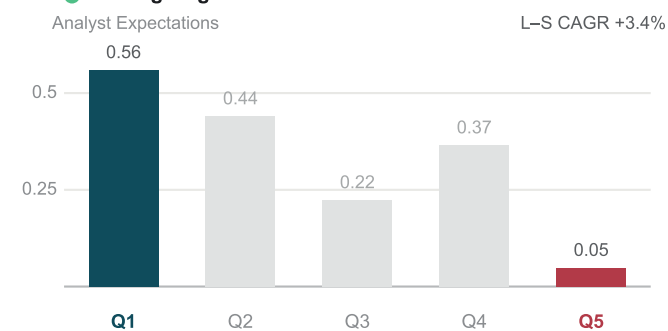
040 ● Share Turnover



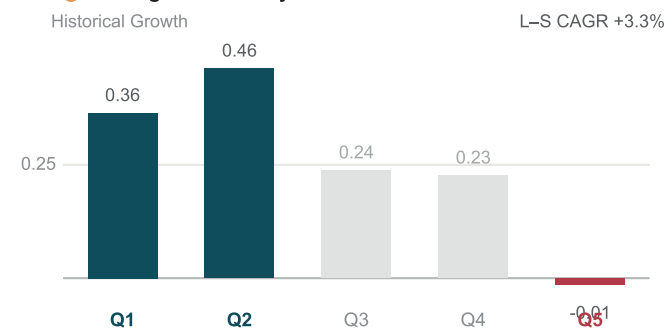
041 ● 1Y Chg in Sales to Price



042 ● 6M Avg Chg 1M Recommendation



043 ● 1Y Chg in Asset Adjusted EPS



044 ● 3Y Chg in Free Cash Flow per Share



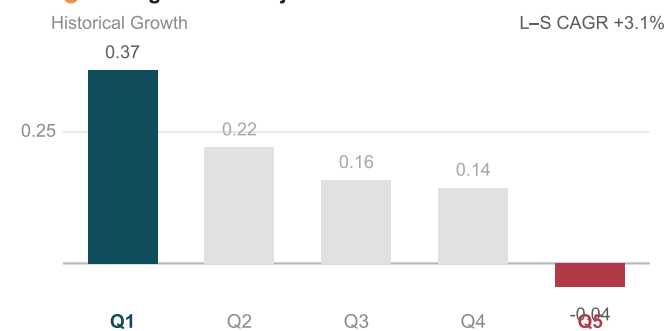
045 ● Capital Acquisition Ratio



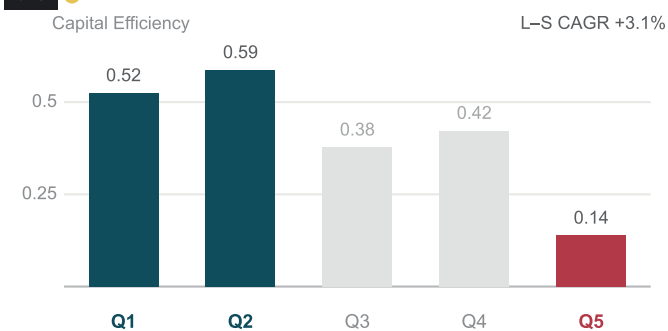
046 ● EBIT Margin



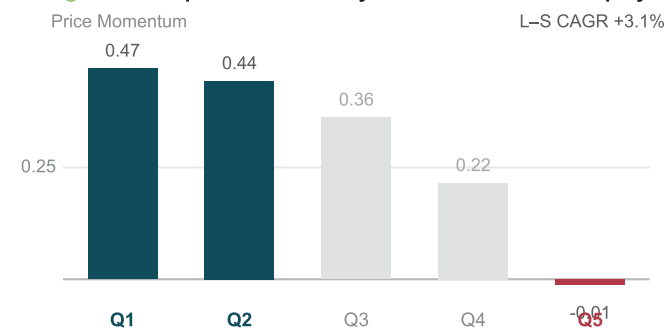
047 ● 1Y Chg in Asset Adjusted Free Cash Flow



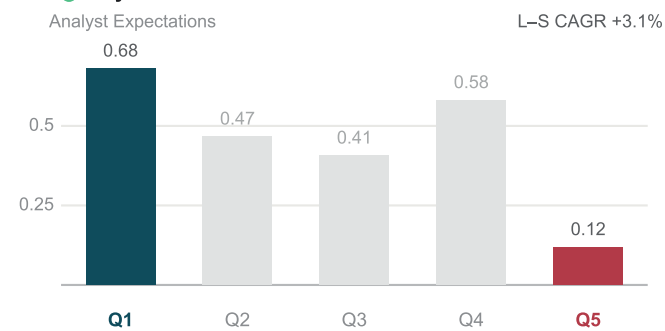
048 ● Receivables Turnover Ratio



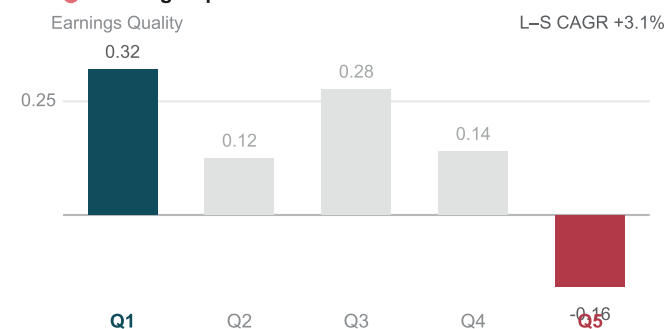
049 ● Ownership Breadth Stability of Institution Ownership by Hol...



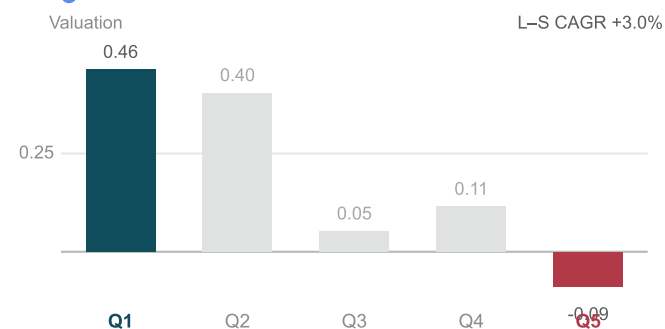
050 ● Buy-to-Sell Recommendation Ratio Less 3M EMA



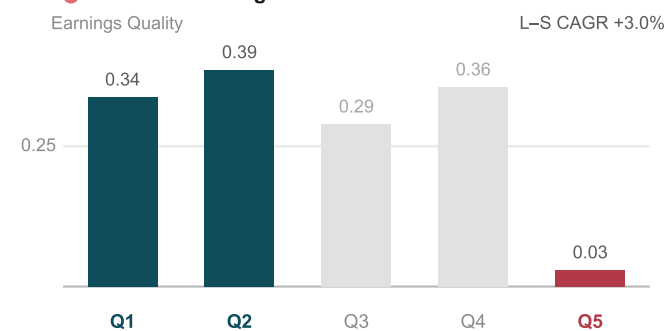
051 ● Working Capital Accruals



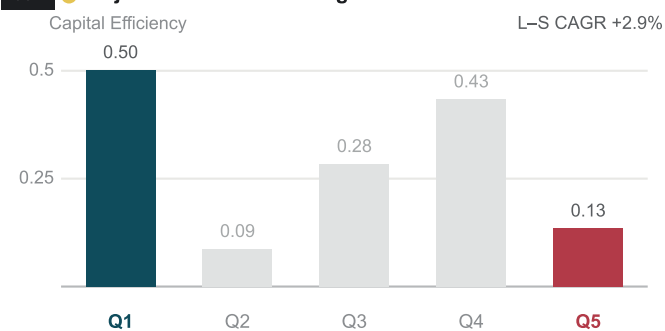
052 ● Assets to Price Ratio

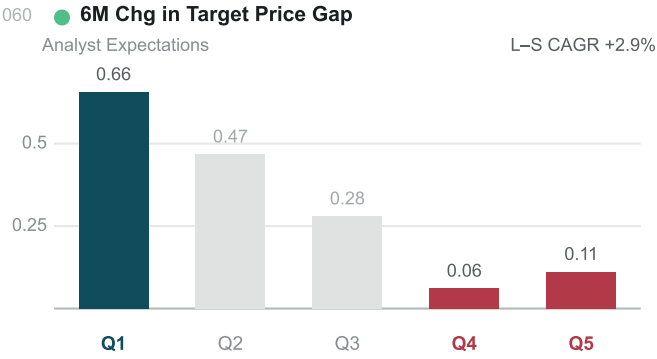
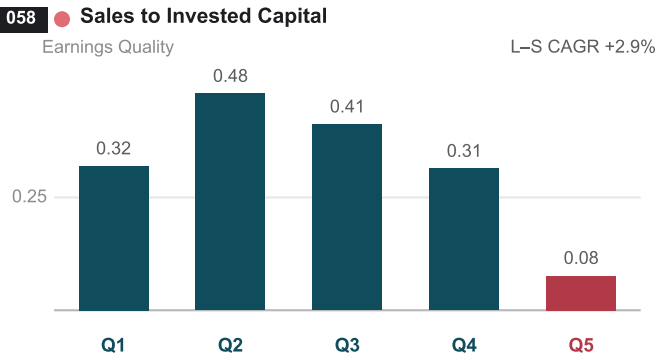
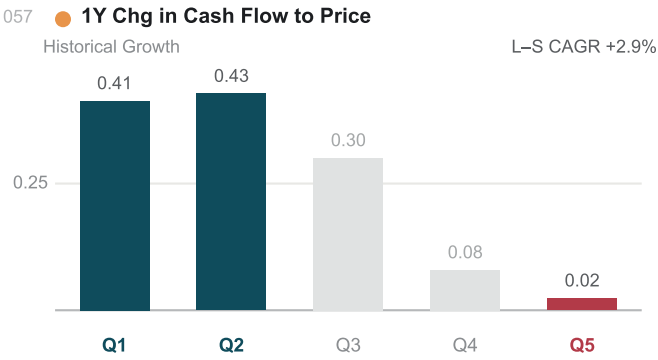
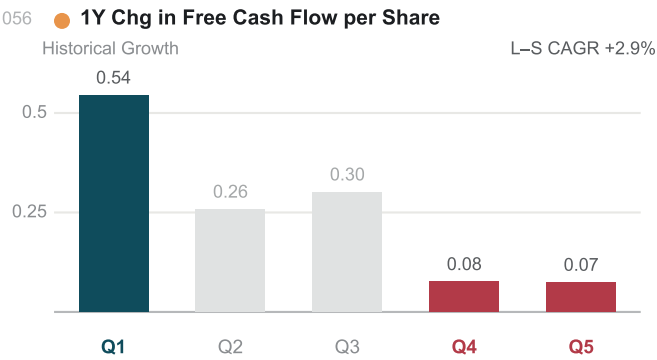
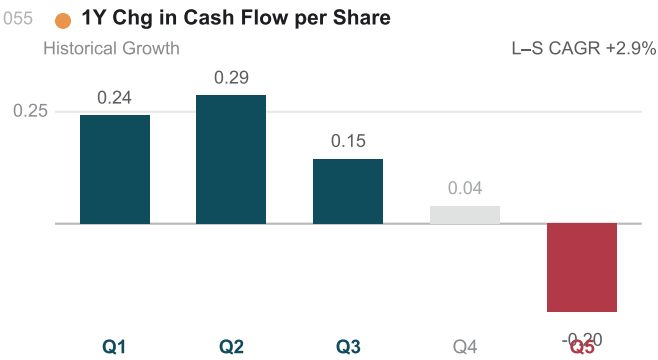


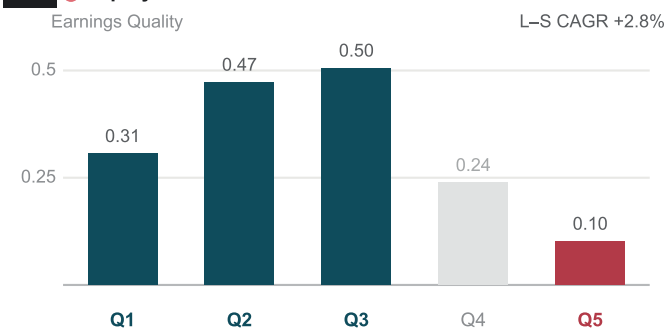
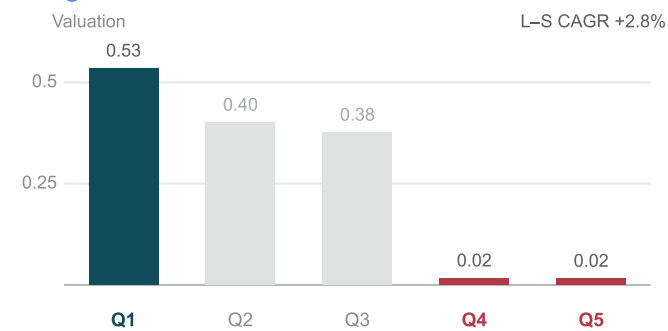
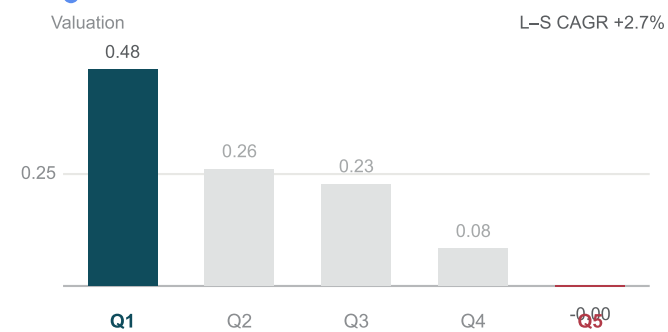
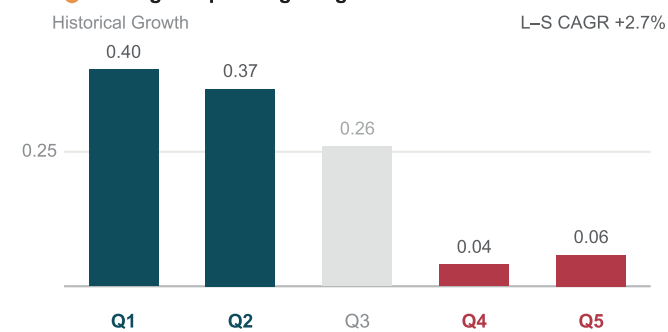
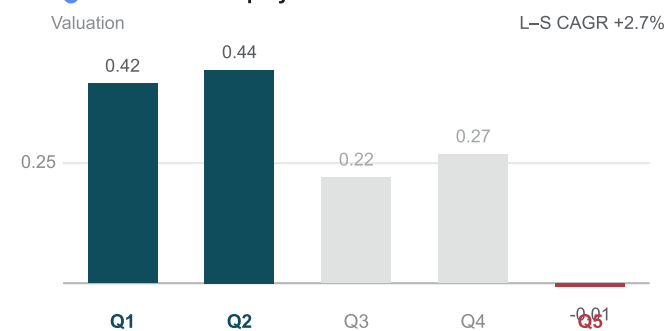
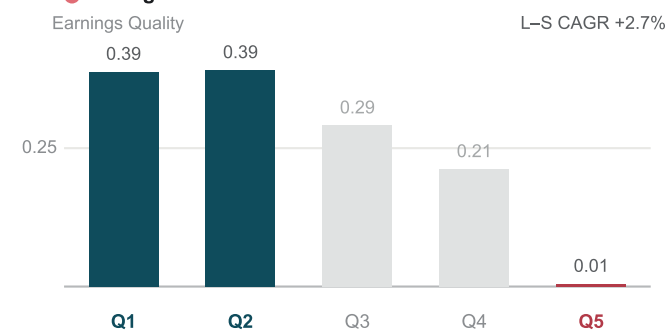
053 ● Gross Profit Margin

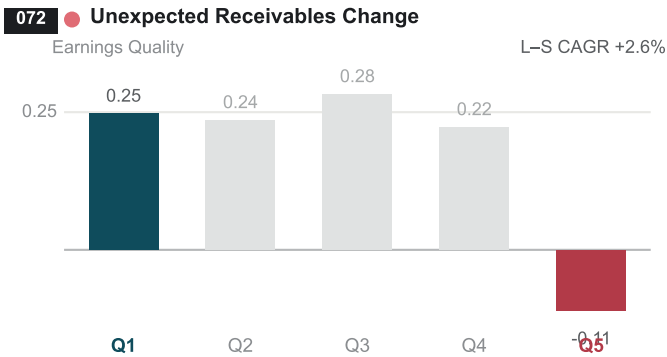
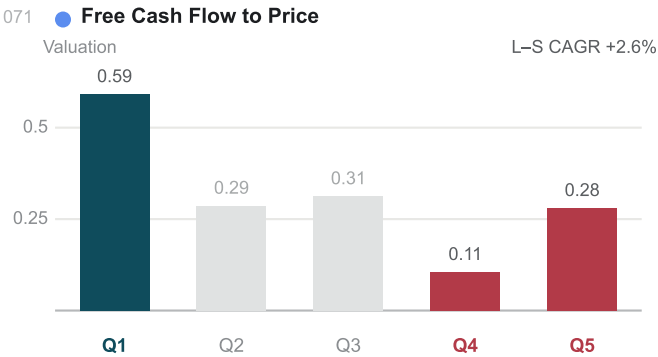
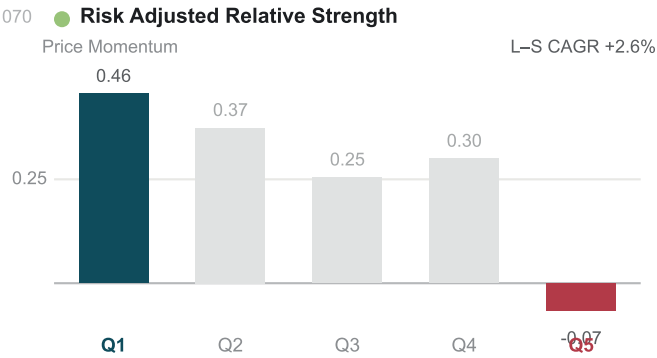
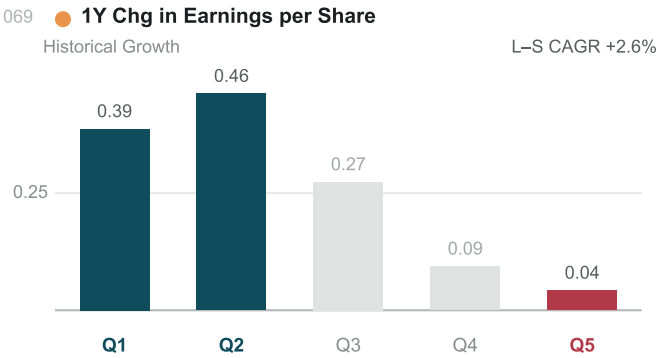
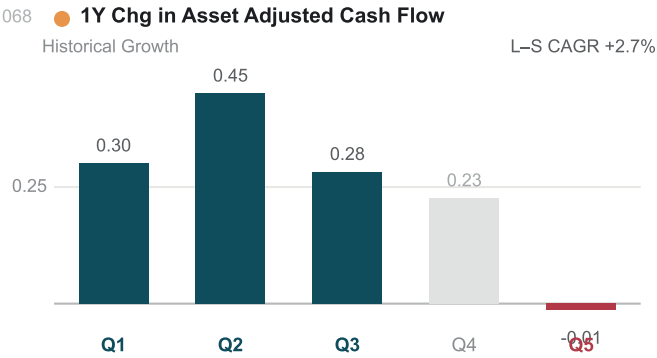
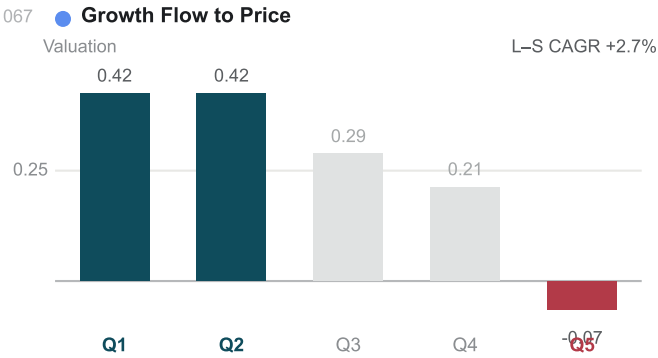


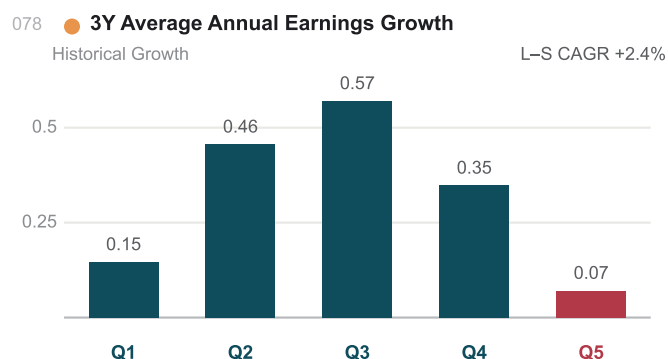
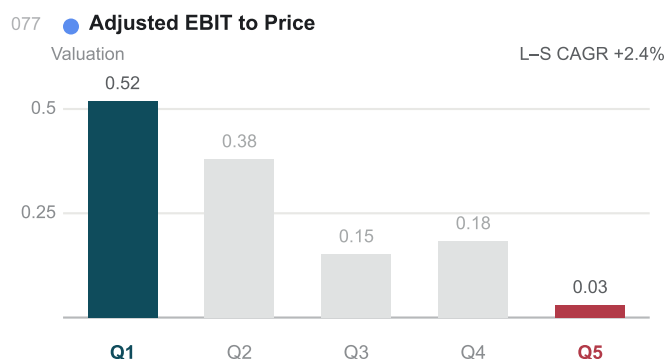
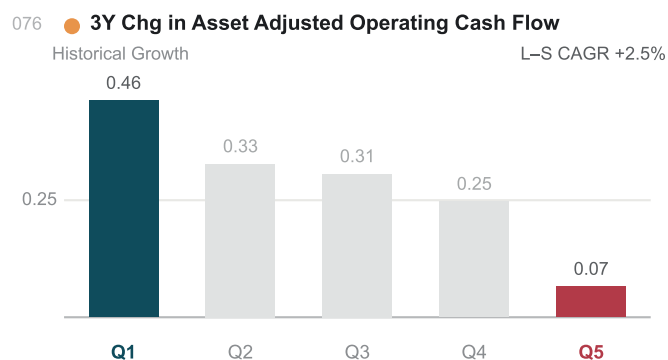
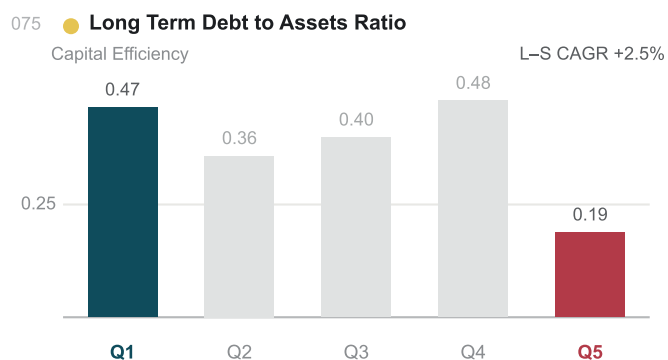
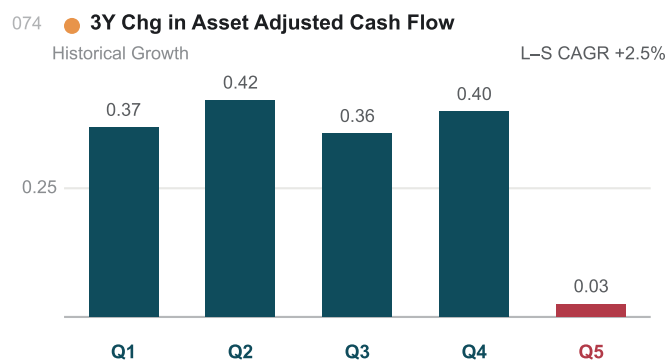
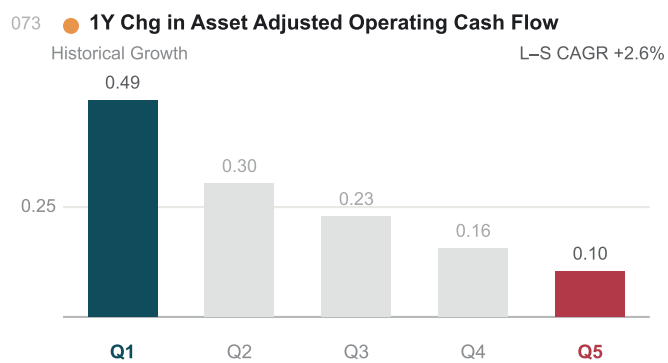
054 ● Adjusted Interest Coverage Ratio



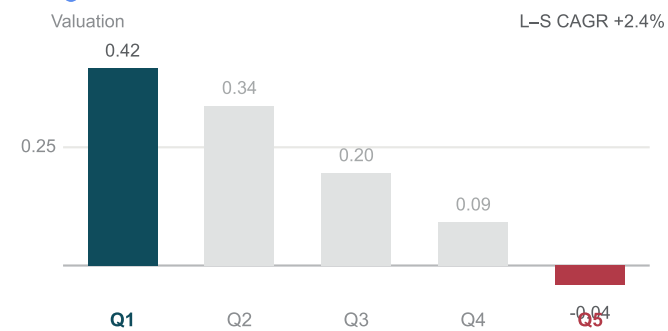


061 Equity Turnover

062 Net Cash Flow to Price

063 Cash to Price

064 1Y Chg in Operating Margin

065 Cash Flow to Equity

066 Change EPS to Sales






079 ● Sales to EV Ratio



080 ● Operating Margin



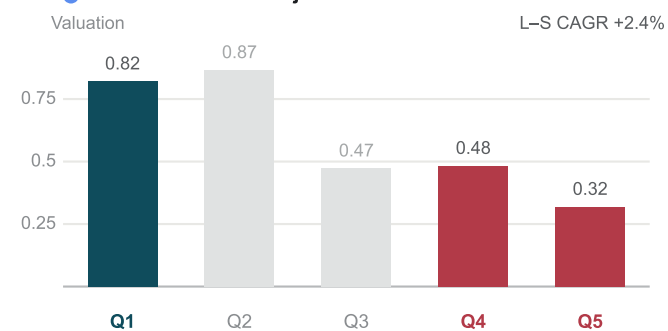
081 ● 3Y Chg in Operating Cash Flow to Price



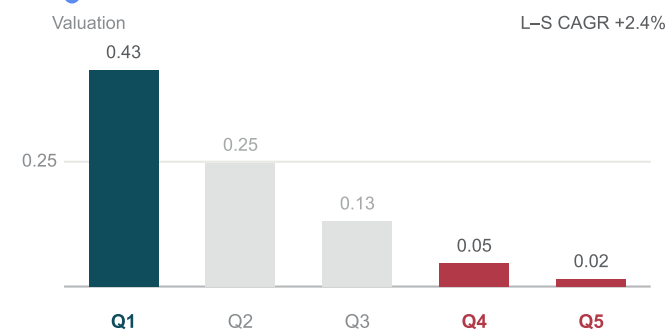
082 ● Retained Earnings to Total Assets

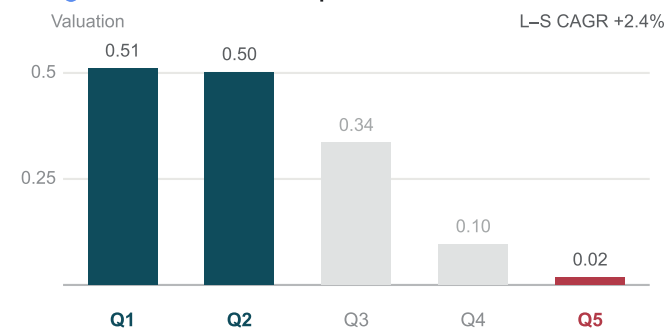
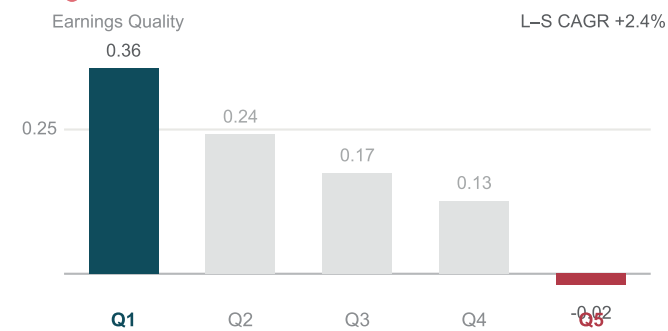
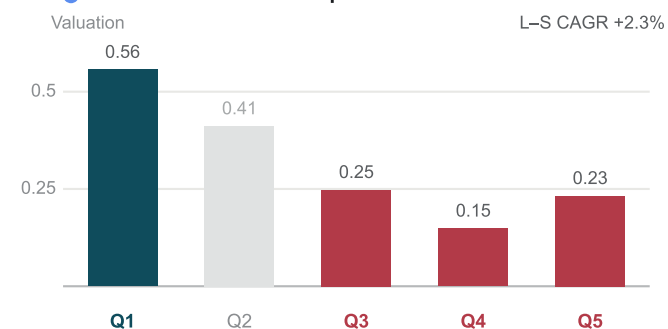
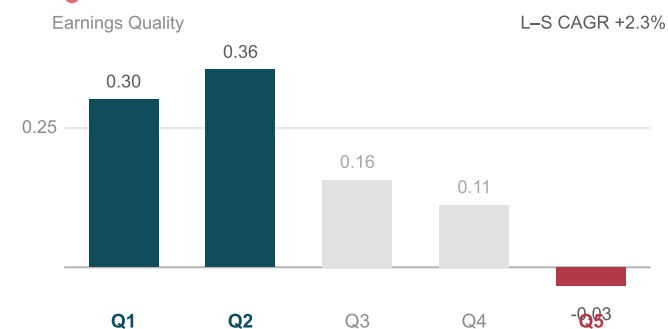
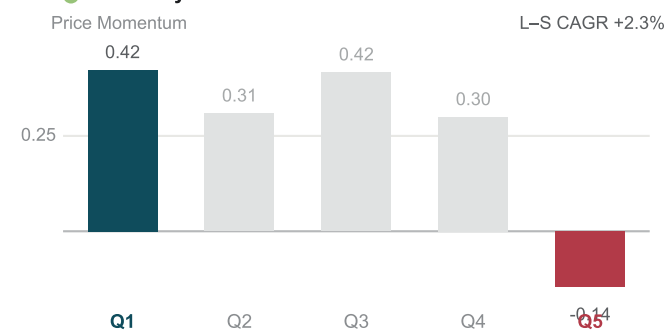
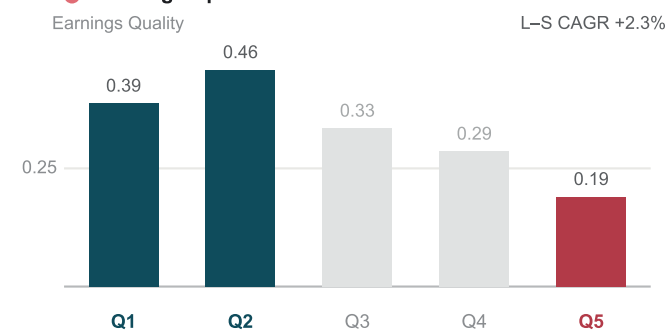


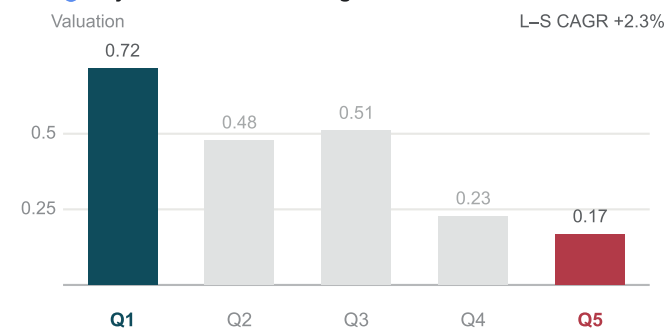
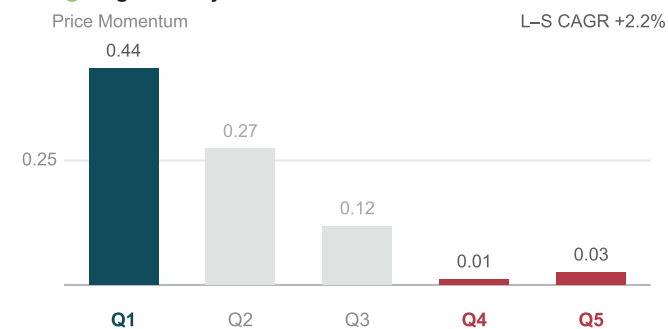
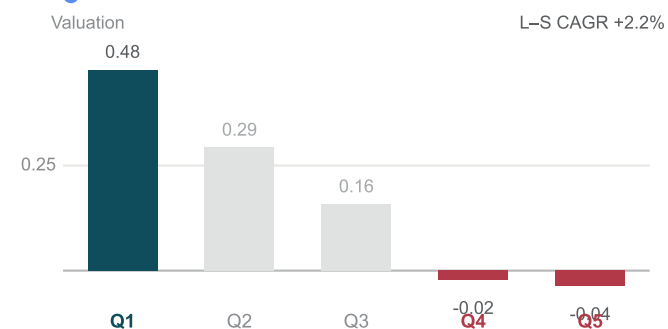
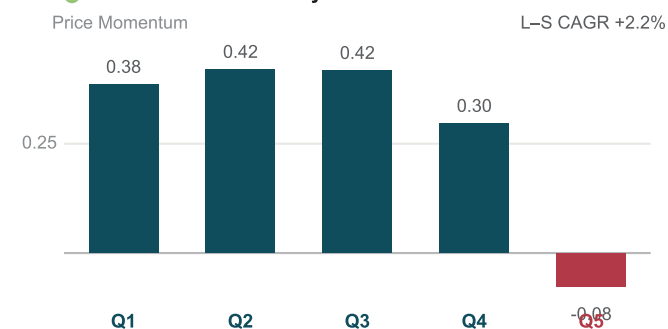
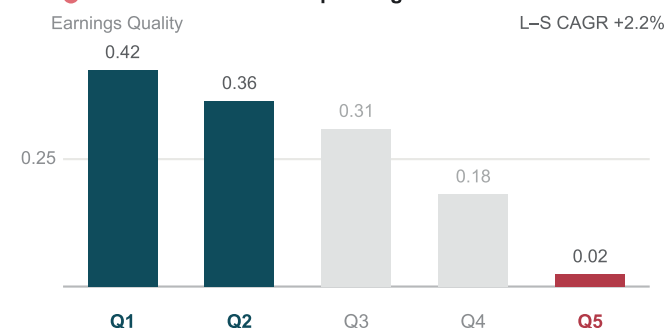
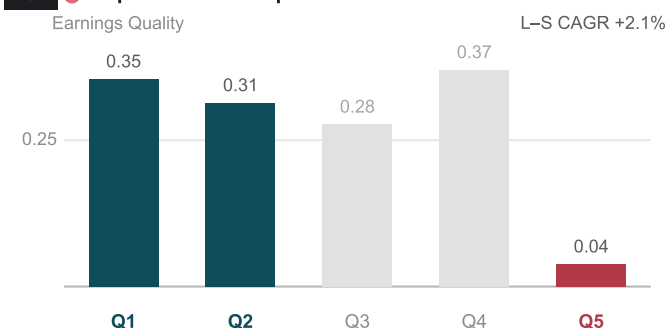
083 ● Inverse P/E Ratio Adj for Growth and Yield



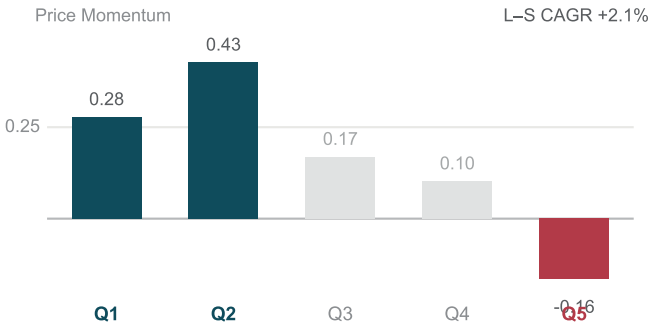
084 ● Cash to EV Ratio



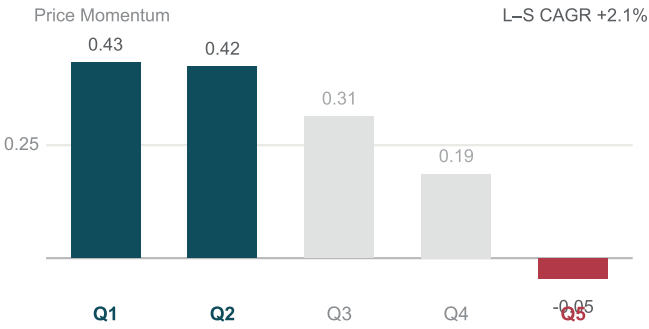
085 ● **Net Cash Flow to Enterprise Value**086 ● **Accrual Ratio - Balance Sheet**087 ● **Free Cash Flow to Enterprise Value**088 ● **Accrual Ratio - Cash Flows**089 ● **Max Daily Return in the Past 6 Months**090 ● **Working Capital to Assets Ratio**

091 ● **Adjusted Forward Earnings to Price**092 ● **Log of Unadjusted Stock Price**093 ● **Book to Price**094 ● **Percent Above 260-Day Low**095 ● **Pretax Income to Net Operating Assets**096 ● **Depreciation to Capex Ratio**

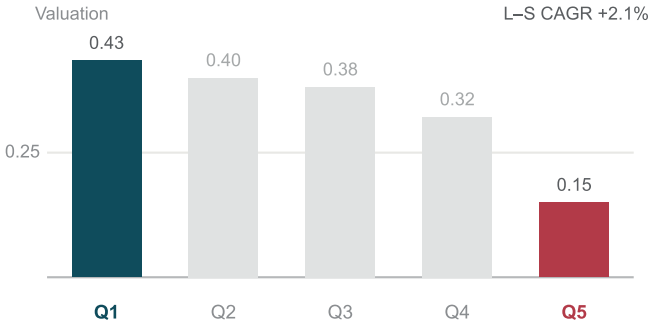
097 ● 15/36 Week Price Ratio



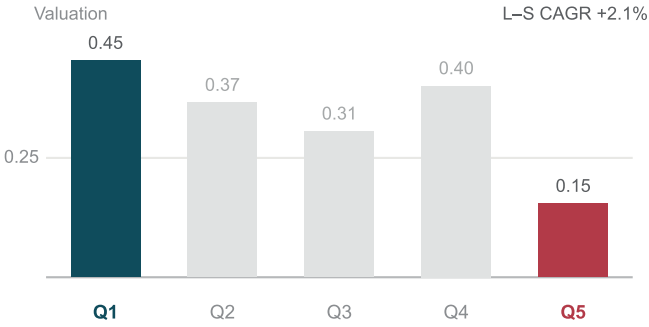
098 ● Adjusted 6 Month Average Share Turn Over



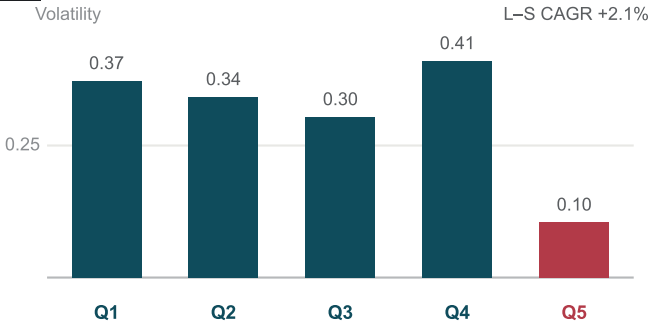
099 ● Solvency Ratio



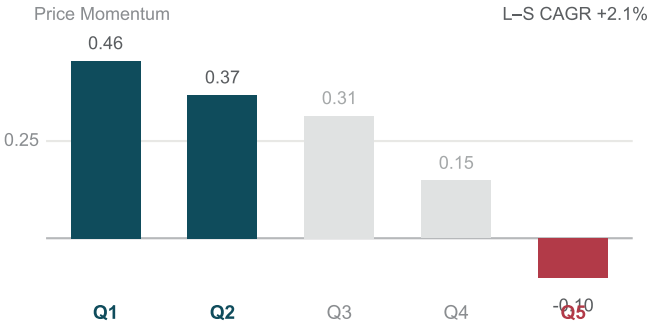
100 ● Cash to Total Assets



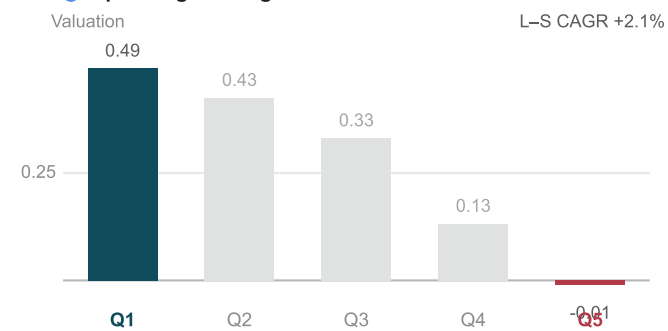
101 ● 60M CAPM Beta



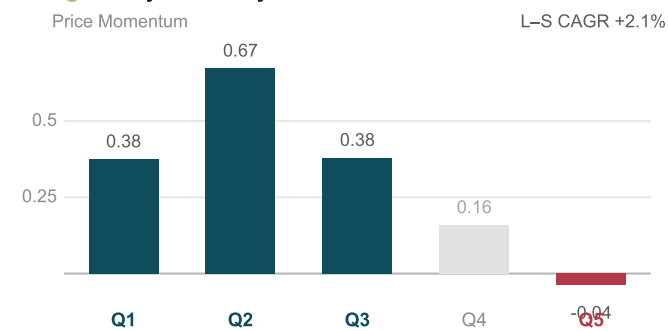
102 ● 12M - 1M Price Momentum



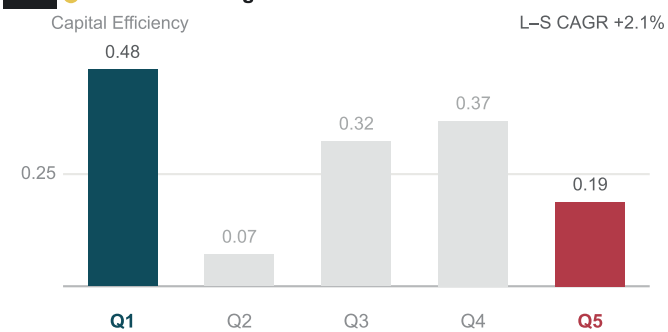
103 ● Operating Earnings to Price Ratio



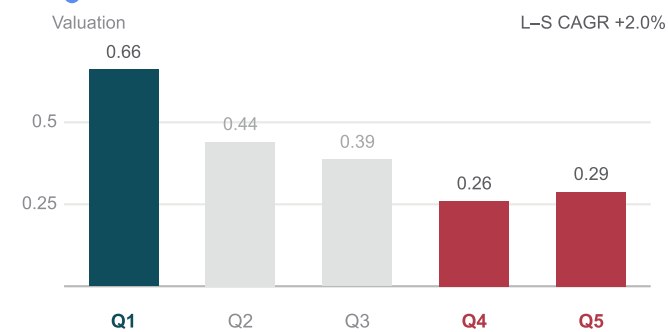
104 ● 50 Day to 200 Day Stock Price



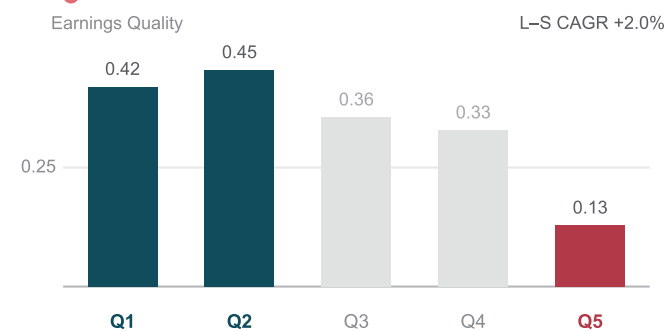
105 ● Interest Coverage Ratio



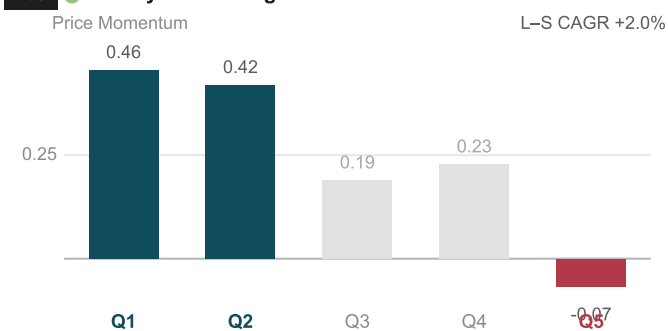
106 ● Forward Free Cash Flow to Price Ratio



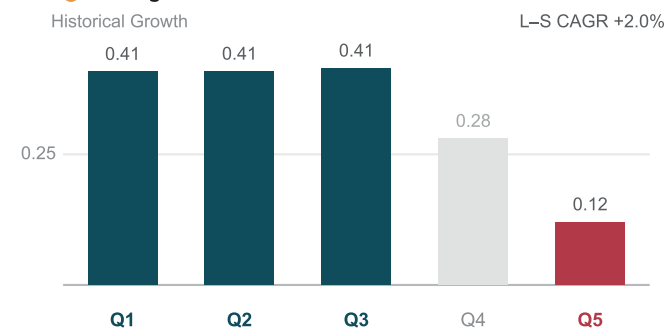
107 ● Cash Flow to Total Asset



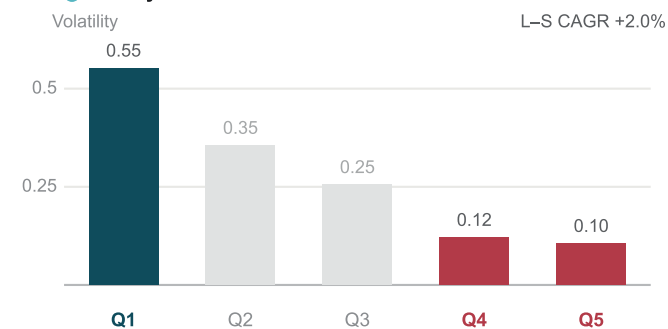
108 ● 50 Day Volume Signal



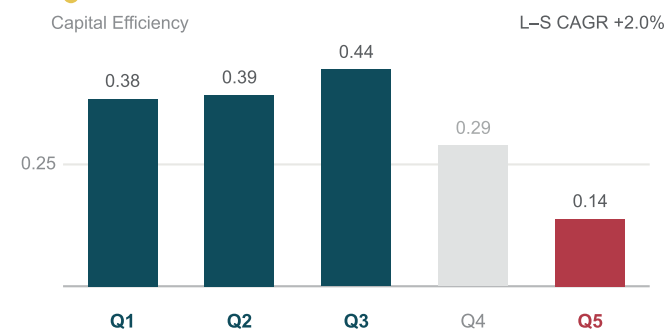
109 ● 3Y Chg in Sales Turnover



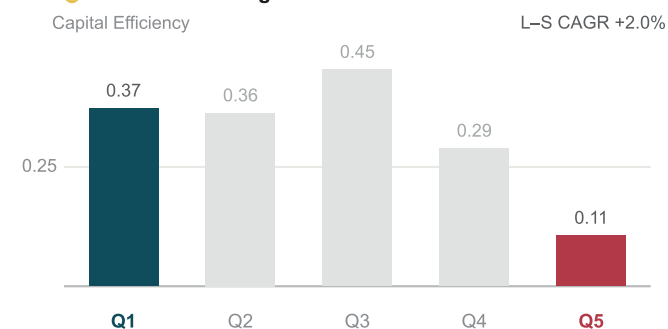
110 ● 90 Day Coefficient of Variation



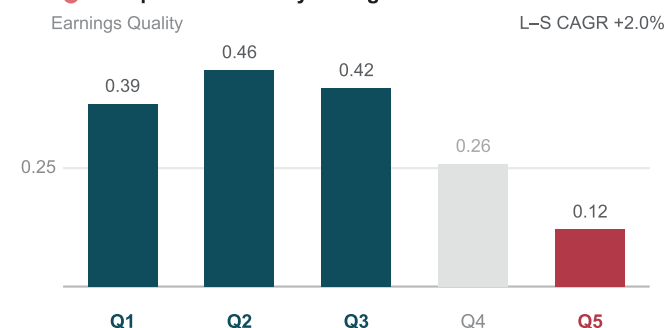
111 ● Debt to Assets Ratio



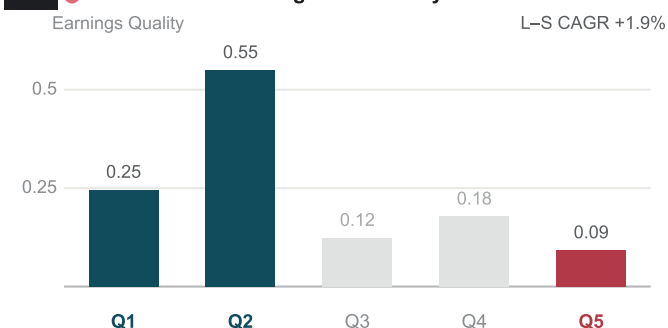
112 ● Pretax Profit Margin



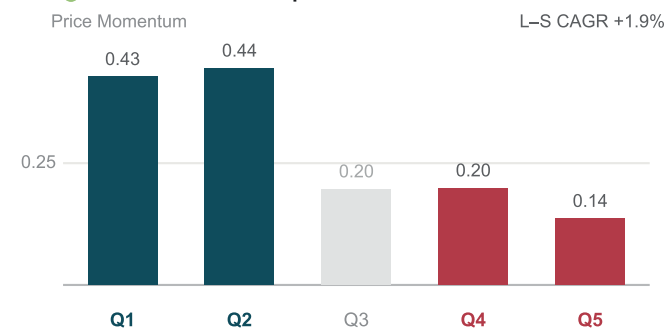
113 ● Unexpected Inventory Change



114 ● Year over Year Change of Inventory to Assets



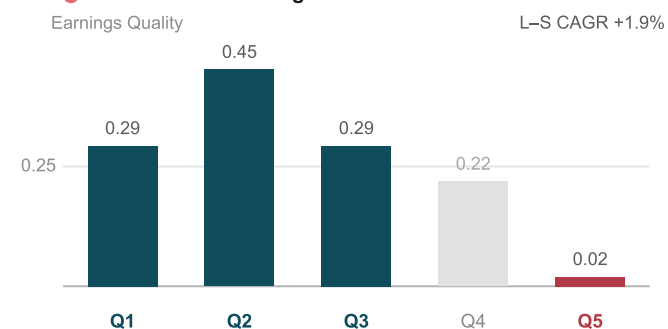
115 ● Institution Ownership Level



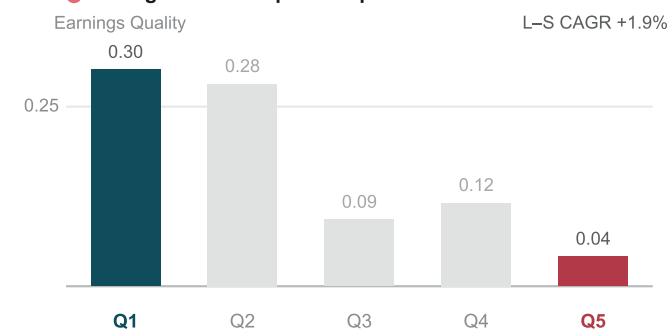
116 ● Dividend Growth



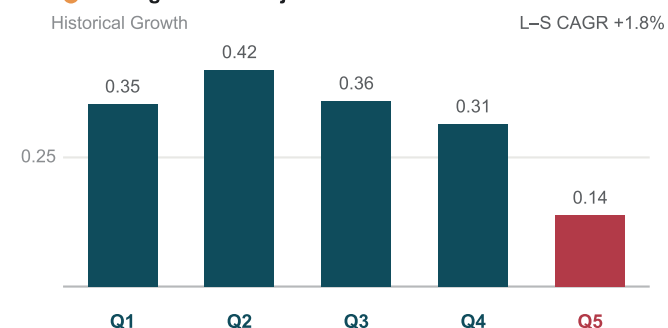
117 ● Year over Year Change of EPS to Sales



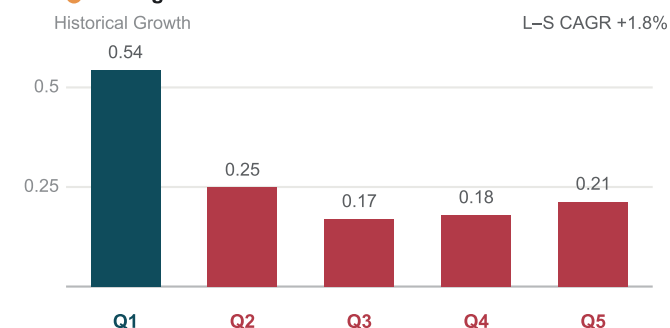
118 ● Change in TTM Depr. to Capex

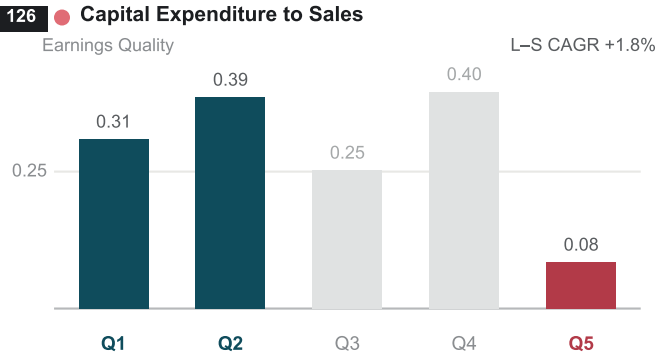
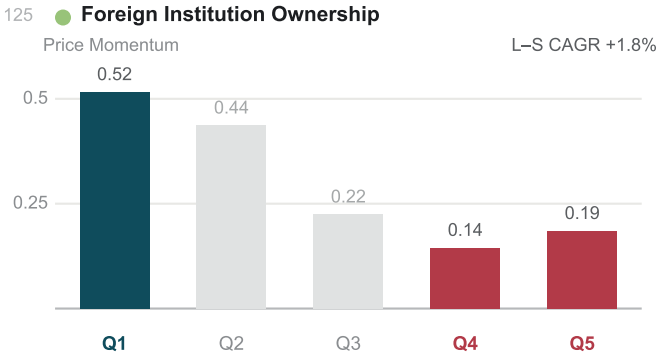
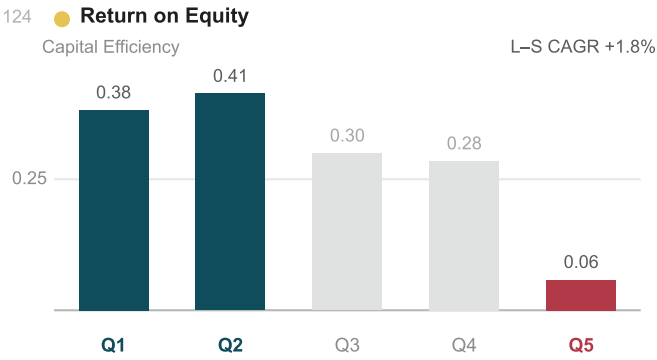
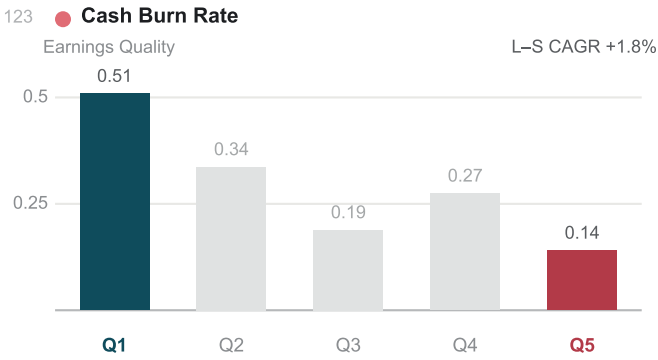
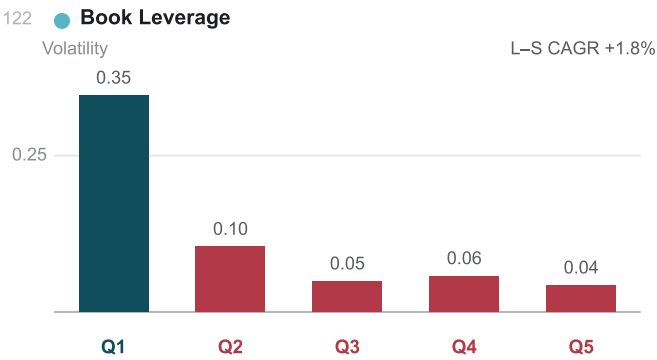
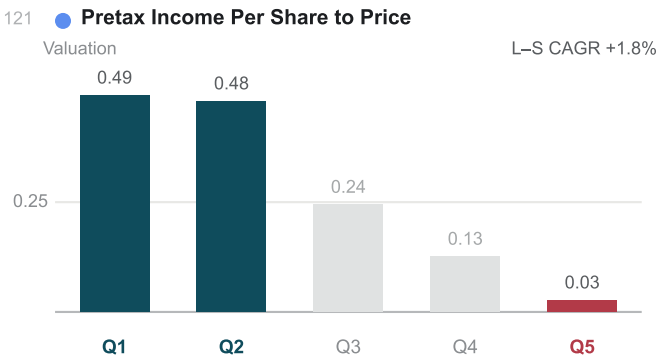


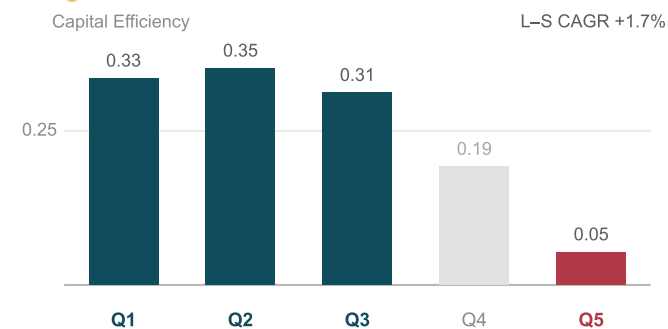
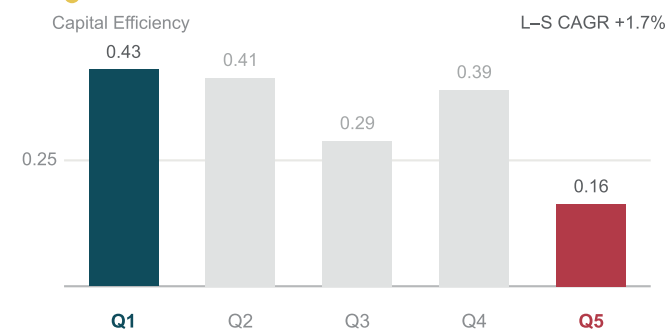
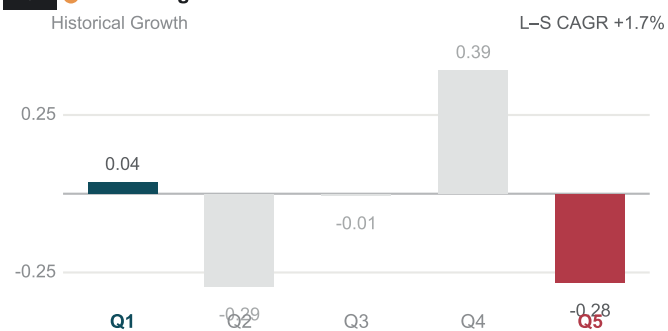
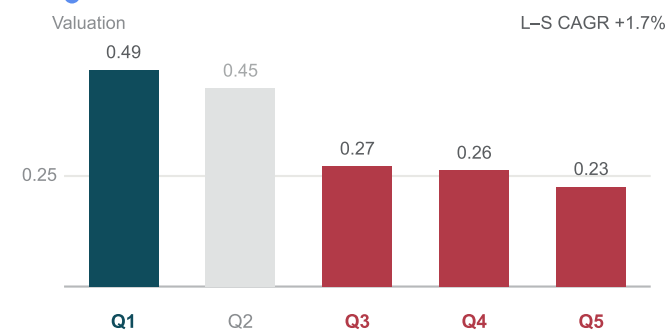
119 ● 3Y Chg in Asset Adjusted EPS

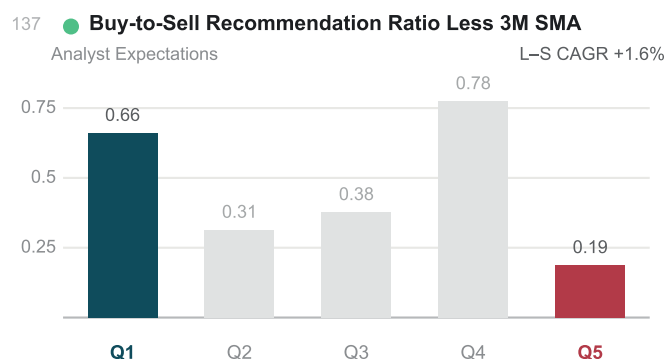
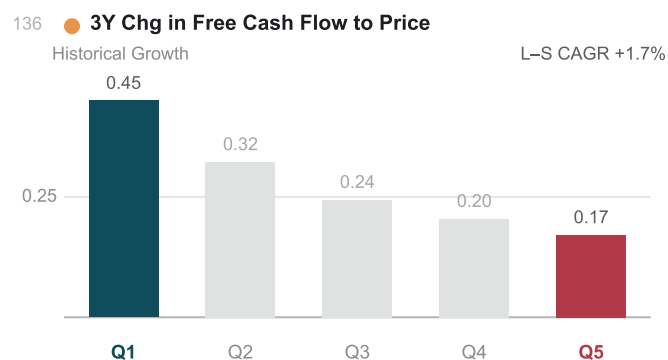
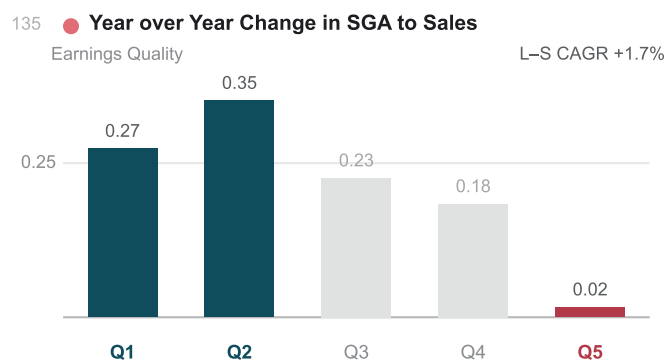
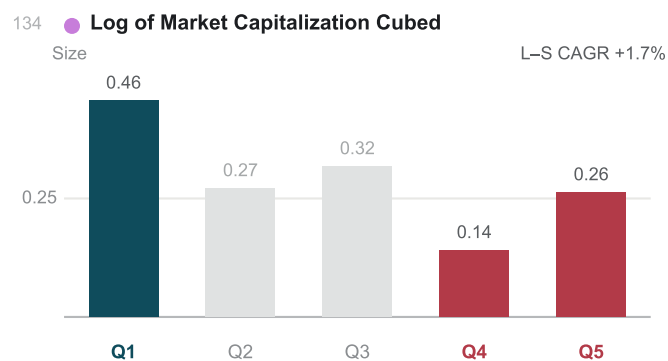
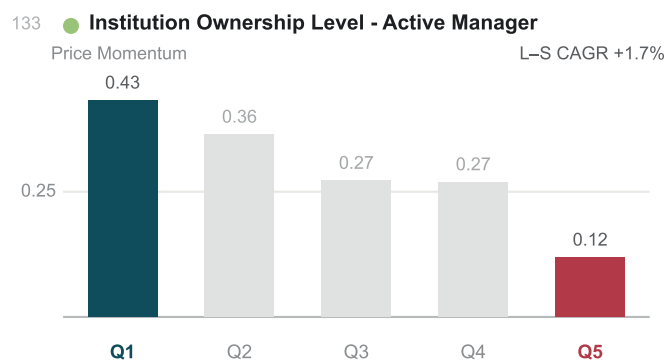


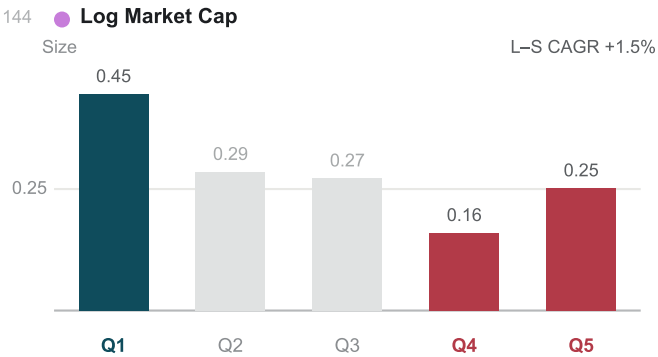
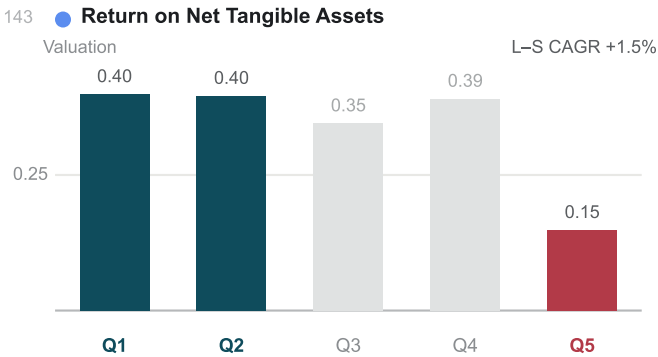
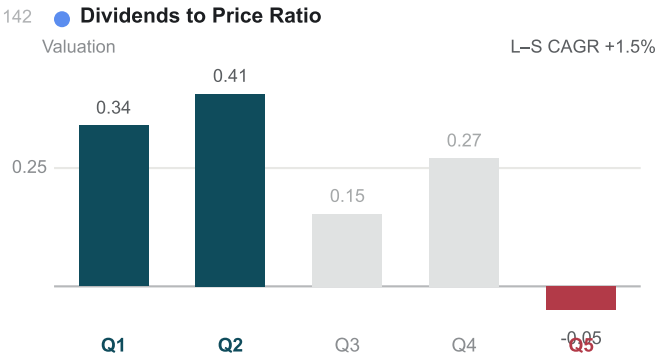
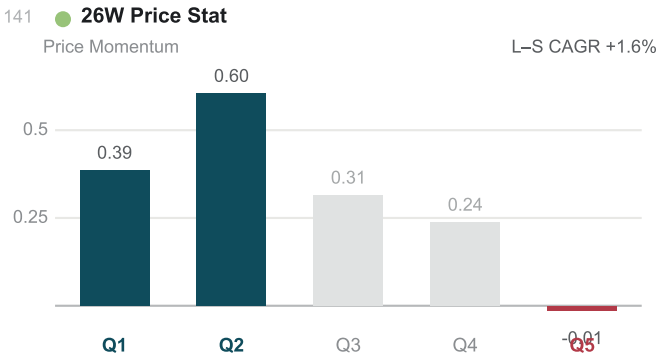
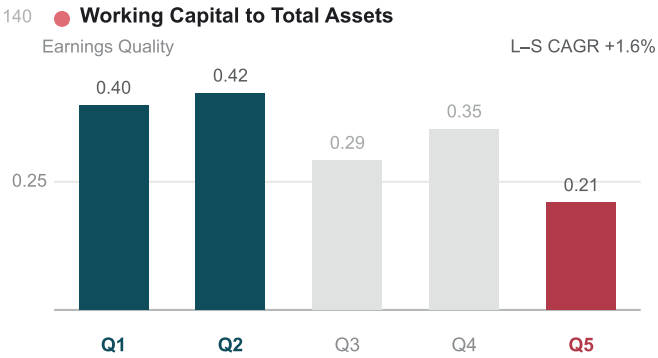
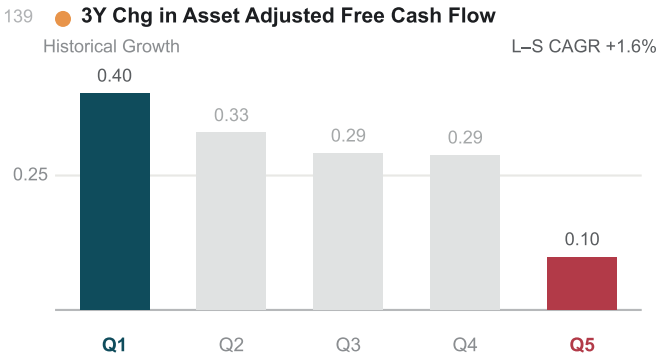
120 ● 3Y Chg in Cash Flow to Price

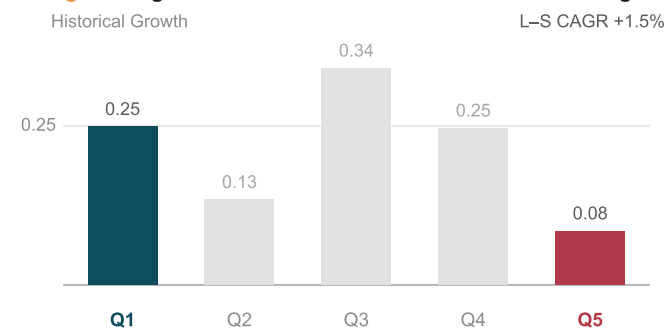
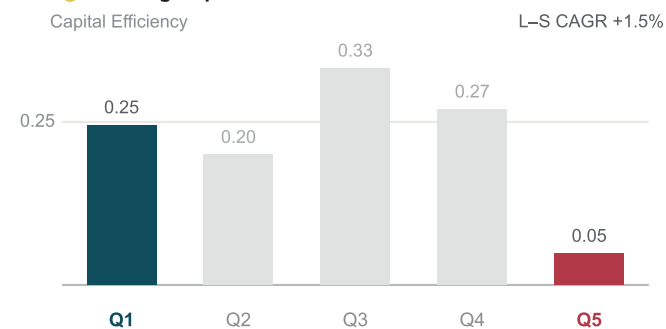
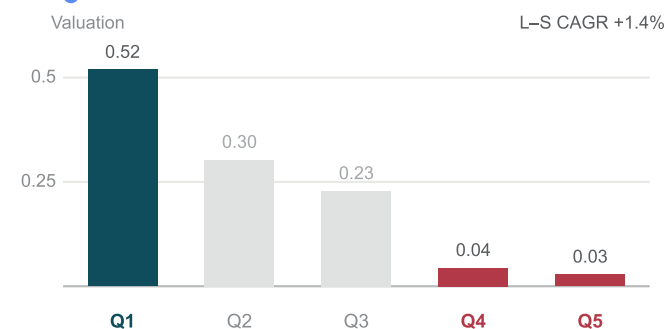
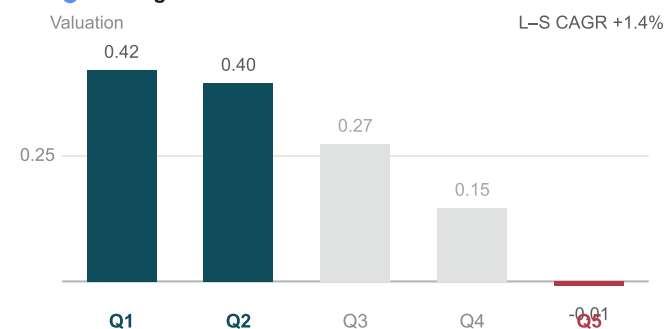
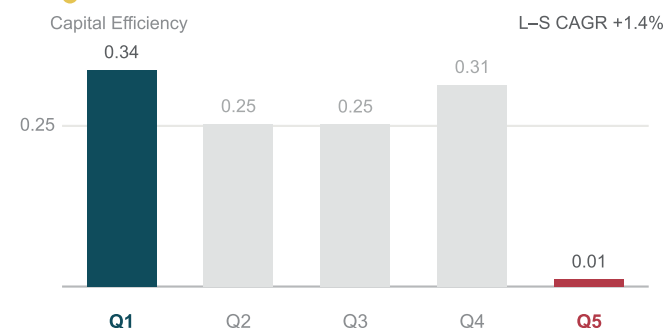




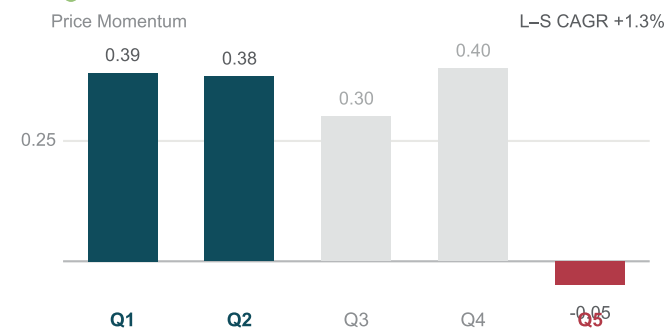
127 ● **Cash Flow to Invested Capital**128 ● **EBITDA to Assets**129 ● **Operating Earnings to Assets Ratio**130 ● **Return on Assets**131 ● **3Y Average Annual Sales Growth**132 ● **Net Current Assets to Price Ratio**



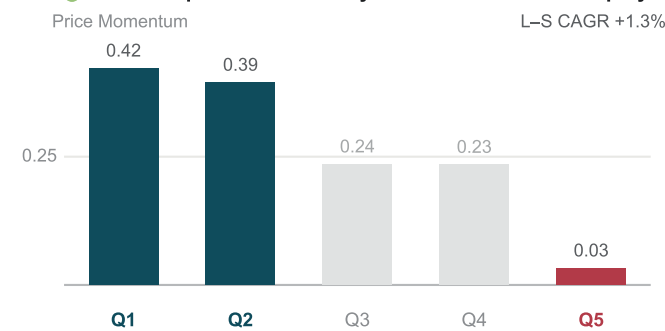


145 ● **Trailing 12-Month Receivables and Inventories/Trailing 12-M...**146 ● **Working Capital Turnover Ratio**147 ● **EBITDA to Price**148 ● **Earnings to Price**149 ● **3Y Chg in Cash Flow per Share**150 ● **Asset Growth - 1 Year**

151 ● 6 Months Price Reversal



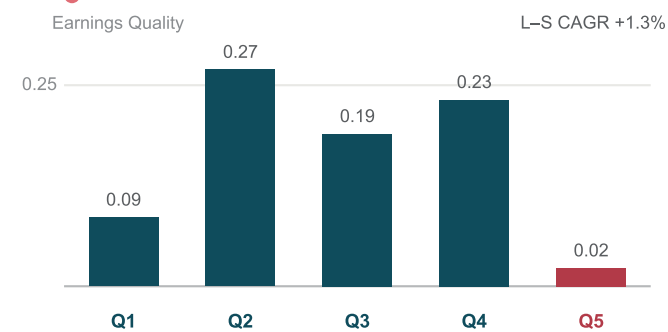
152 ● Ownership Breadth Stability of Institution Ownership by Buyers



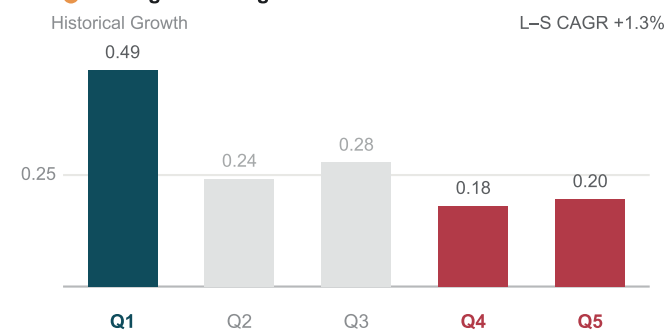
153 ● Forward Earnings to Price



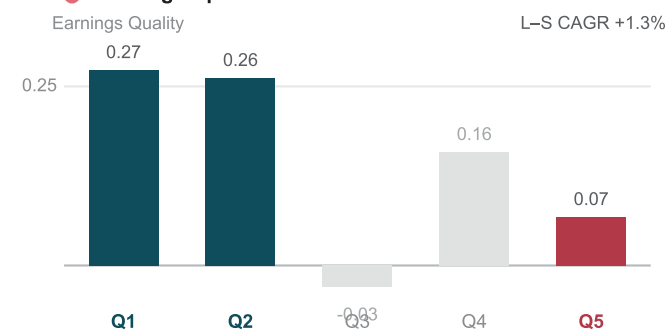
154 ● Cash Ratio

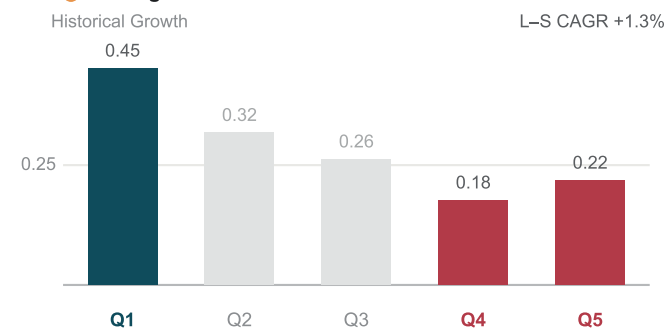
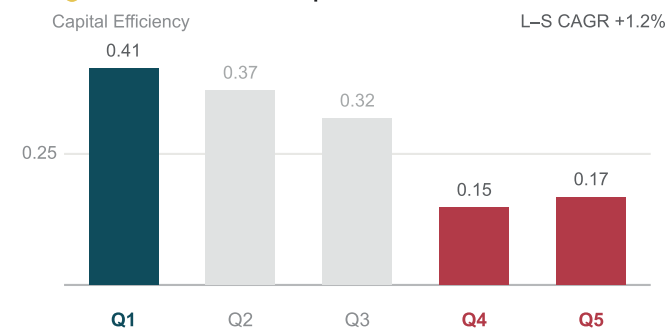
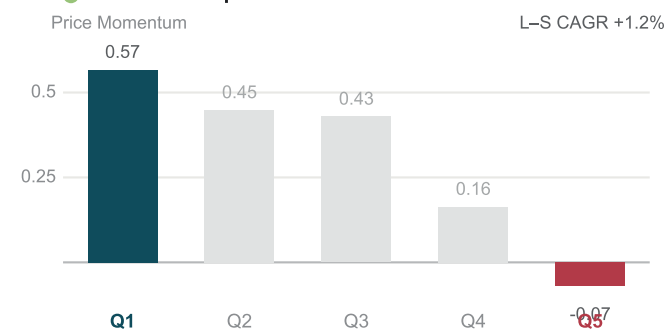
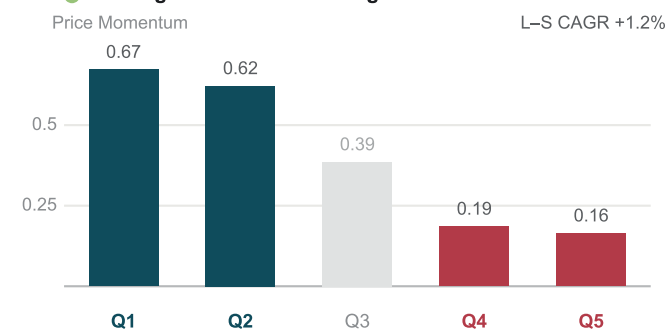
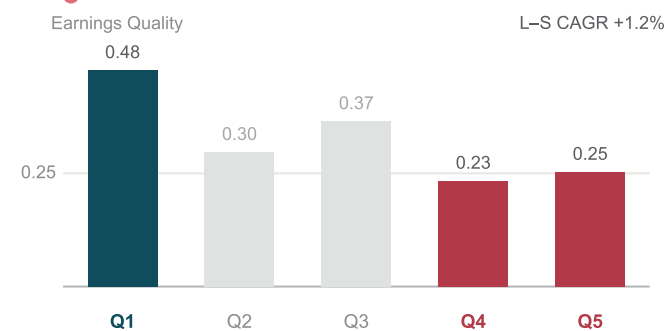
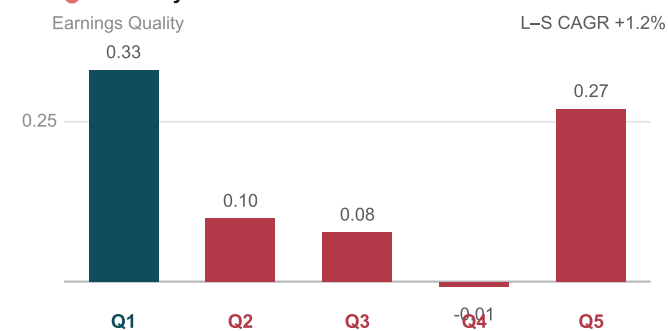


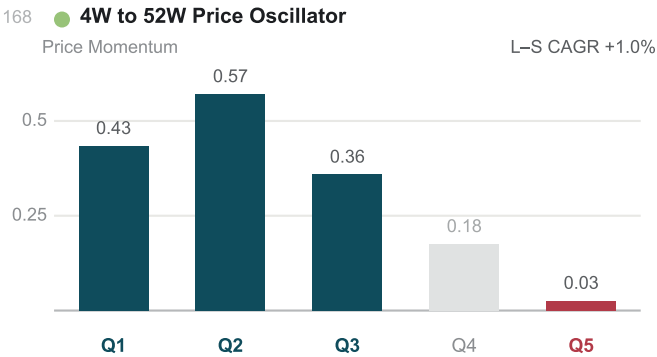
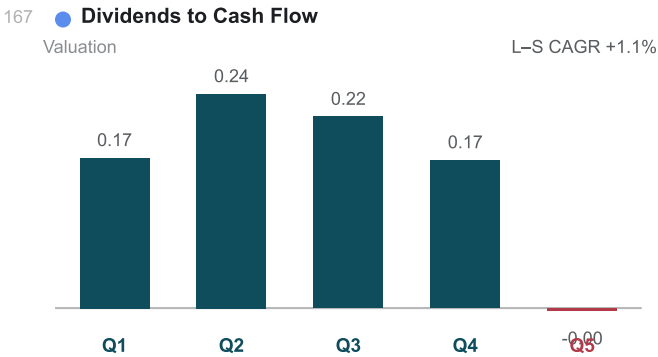
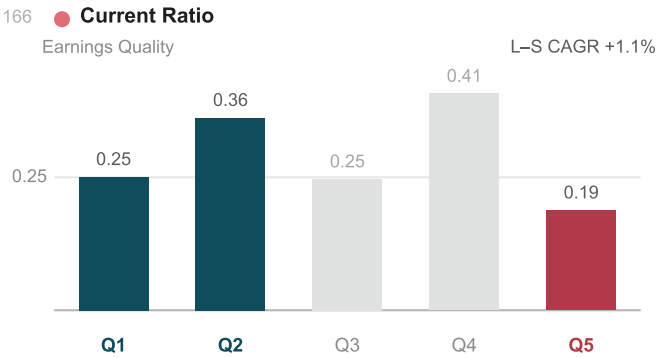
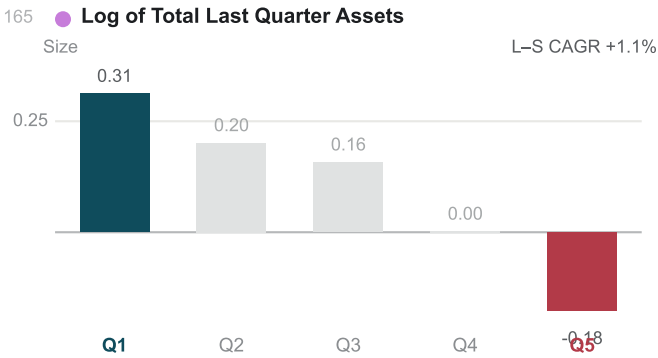
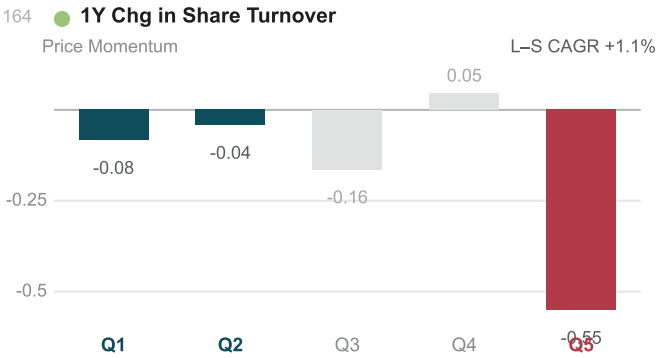
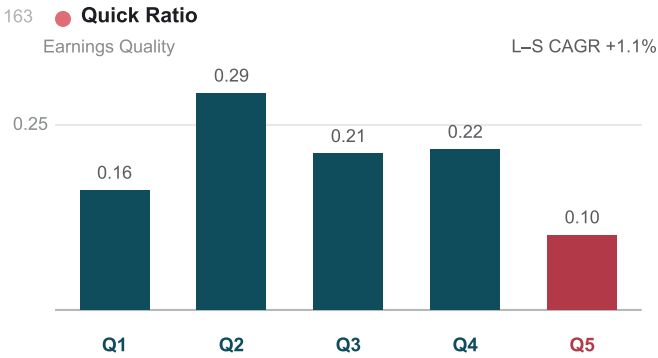
155 ● 3Y Chg in Earnings to Price

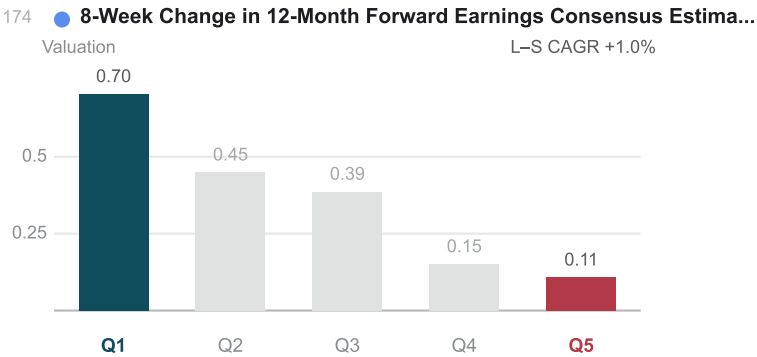
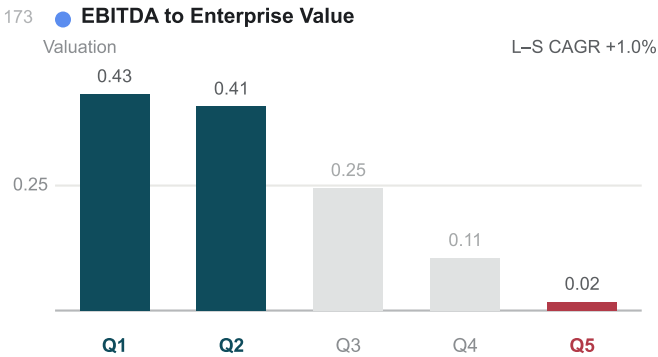
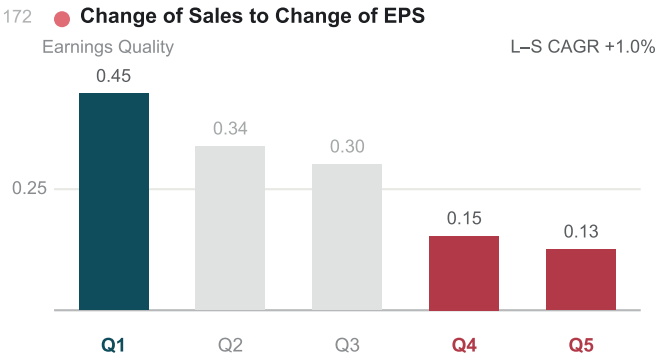
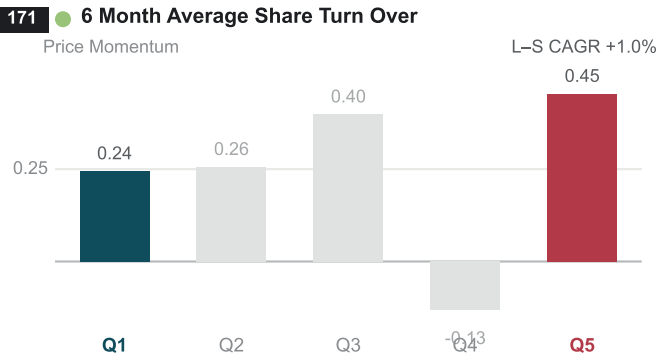
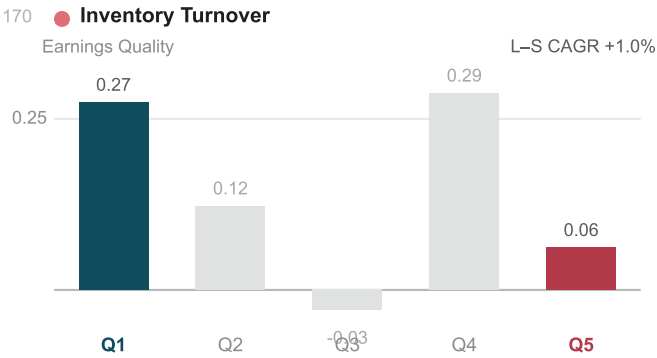
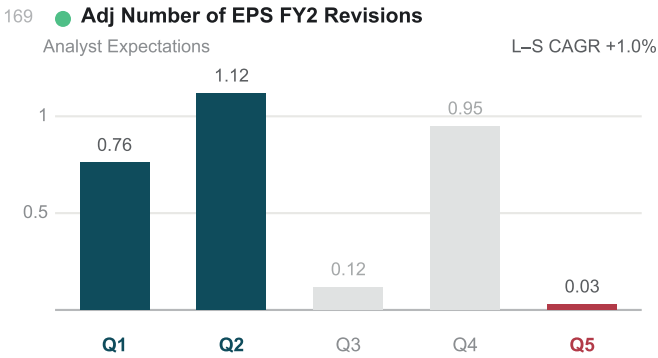


156 ● Working Capital to Sales

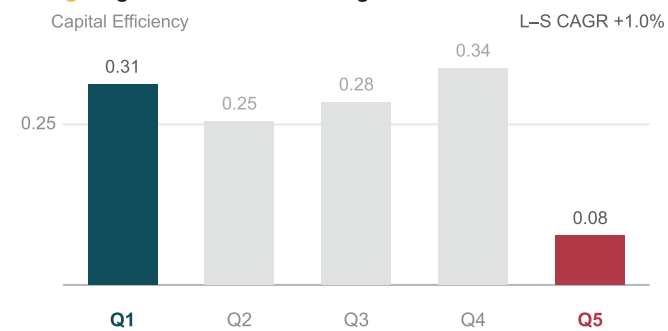


157 ● **3Y Chg in Sales to Price**158 ● **Return on Invested Capital**159 ● **60M CAPM Alpha**160 ● **Closing Price to 52 Week High**161 ● **Cash to Sales**162 ● **Inventory to Assets Ratio**

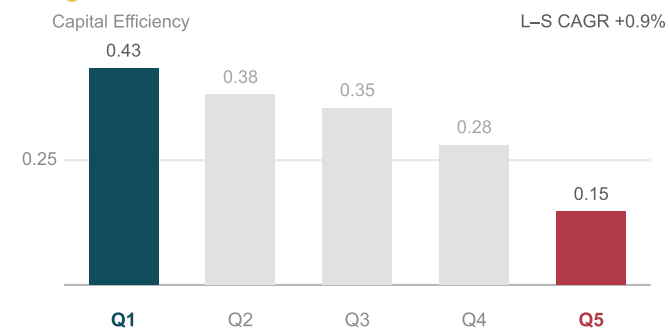




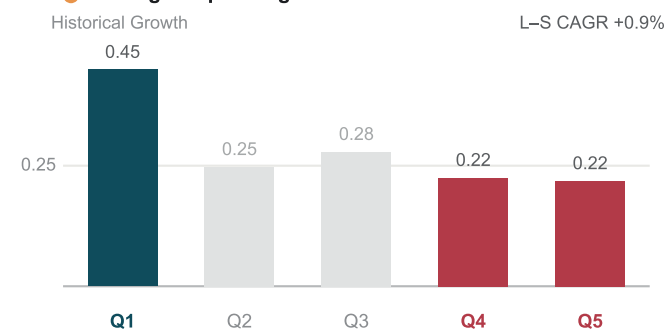
175 ● Degree of Financial Leverage



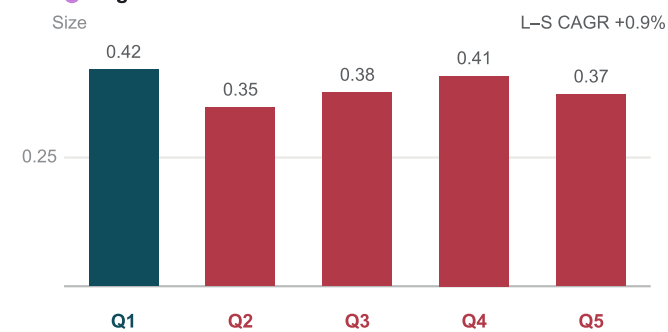
176 ● EBIT to Assets



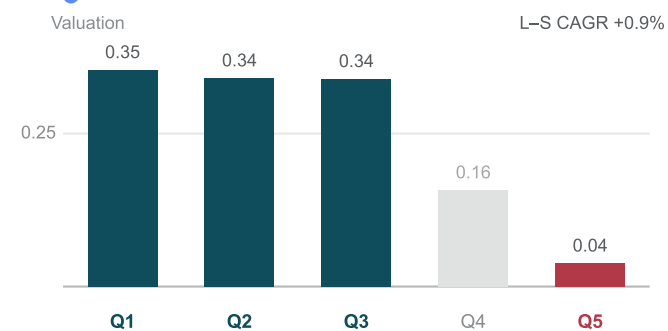
177 ● 1Y Chg in Operating Cash Flow to Price



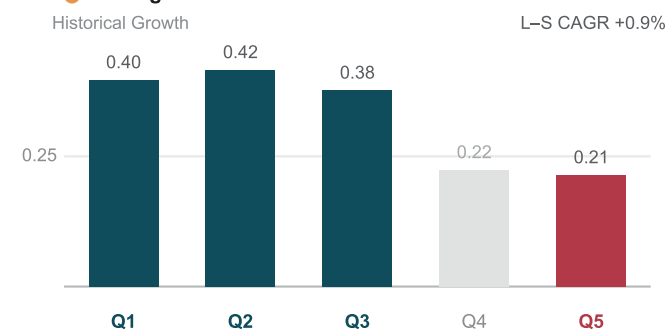
178 ● Log TTM Sales

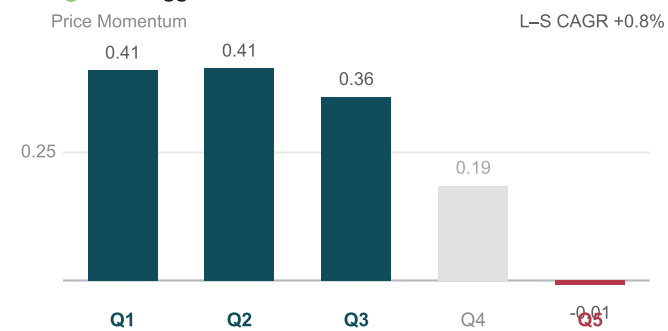
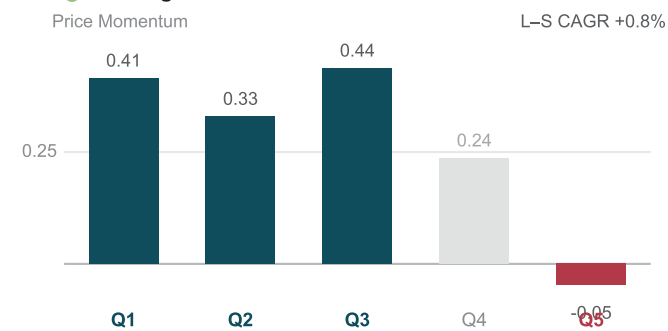
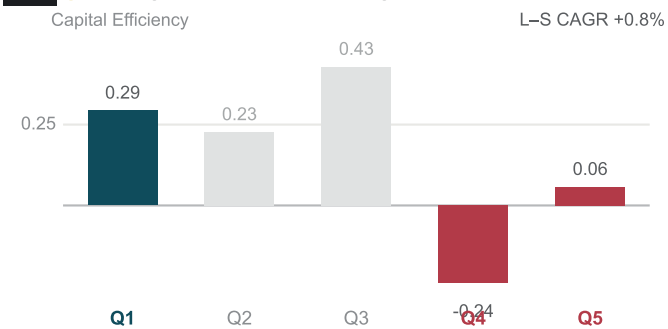
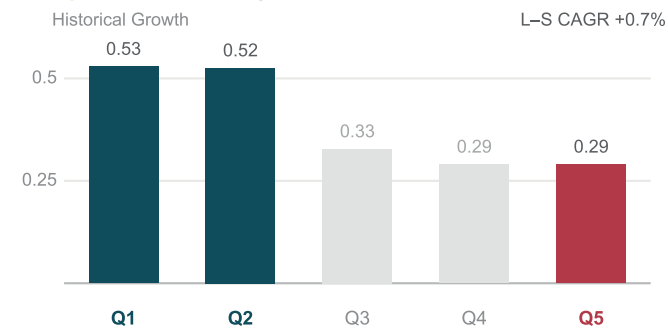
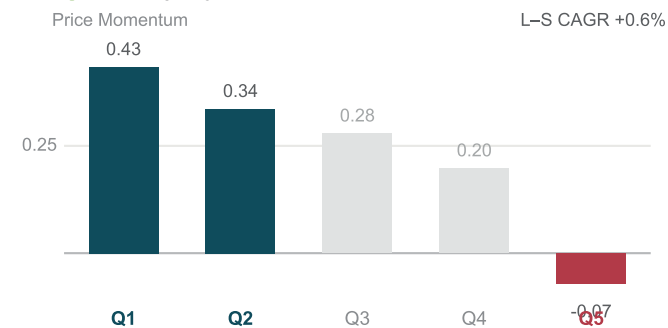


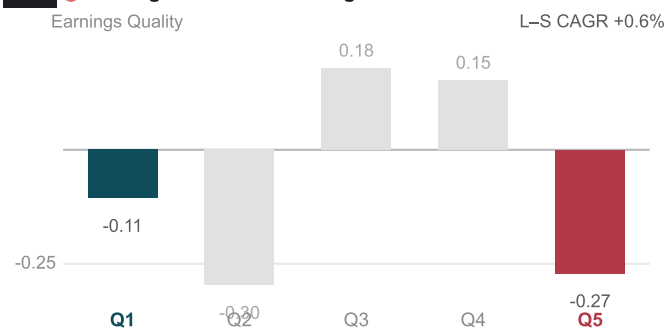
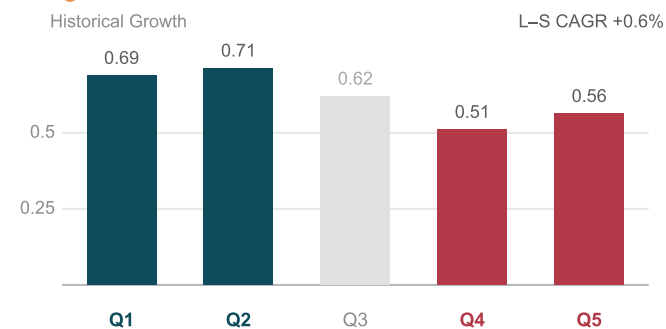
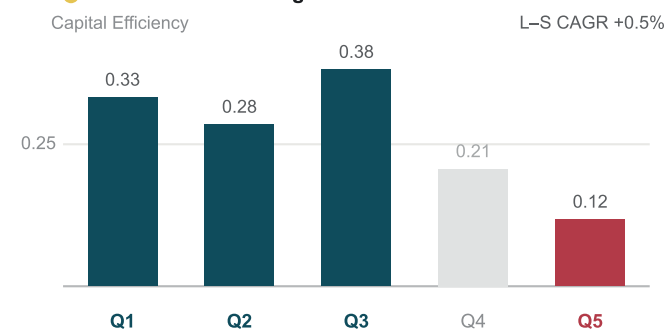
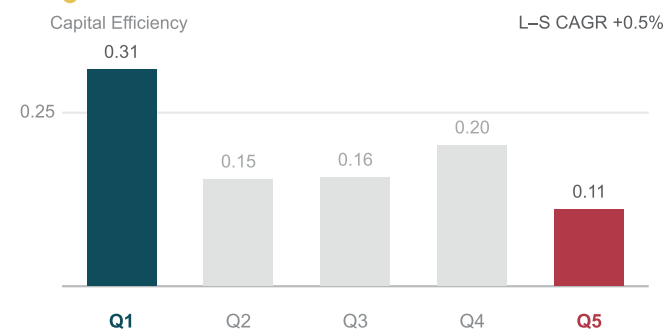
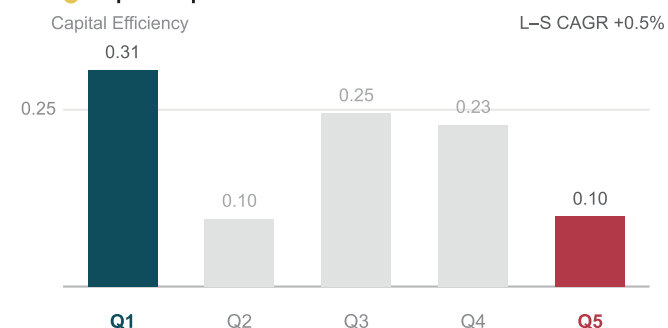
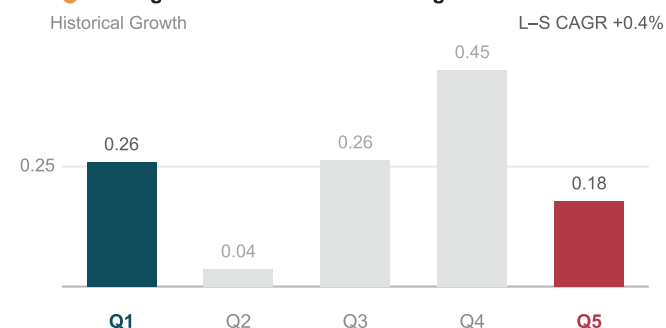
179 ● Normalized EP

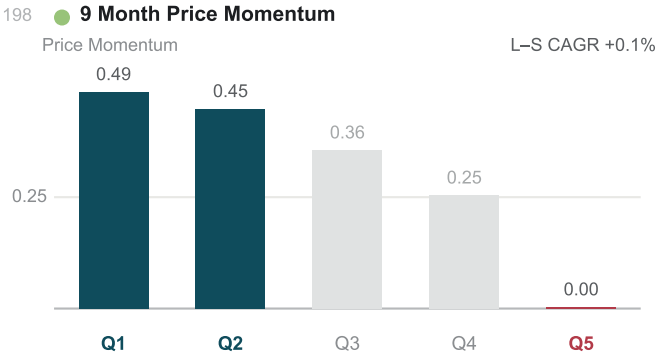
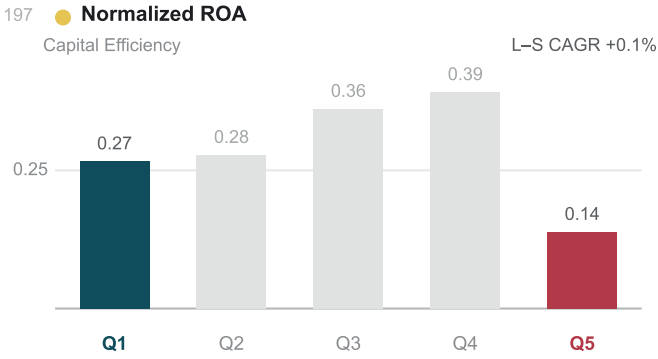
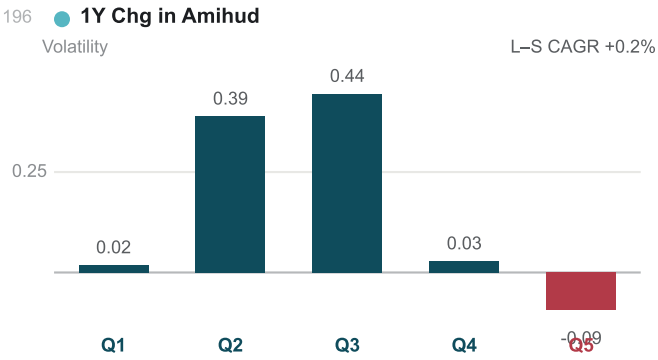
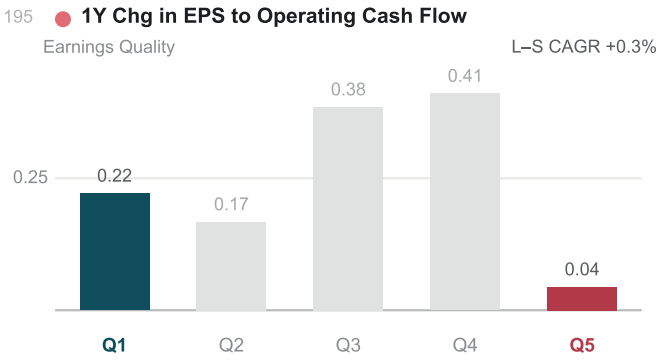
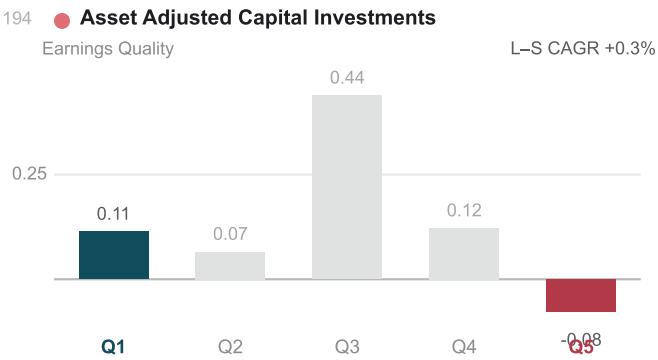
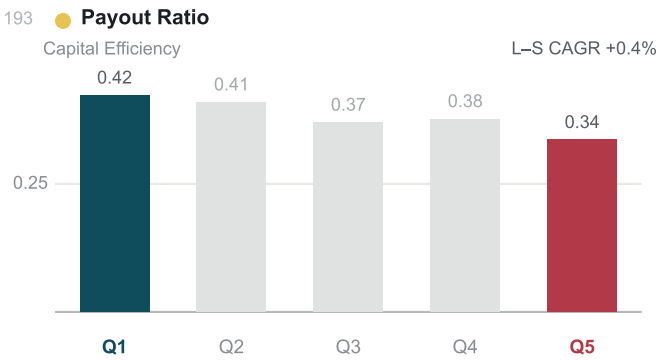


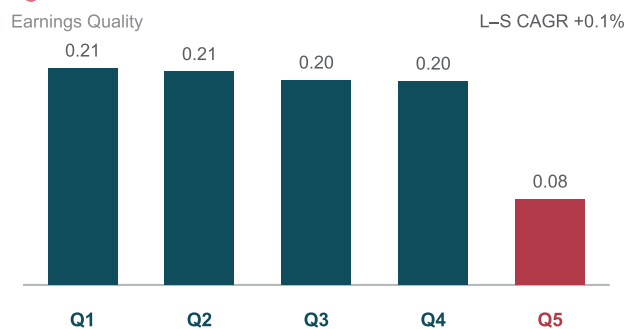
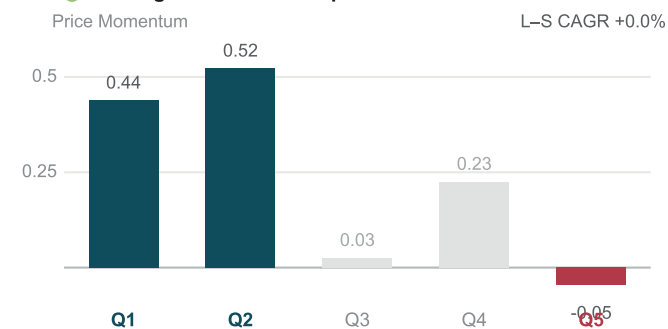
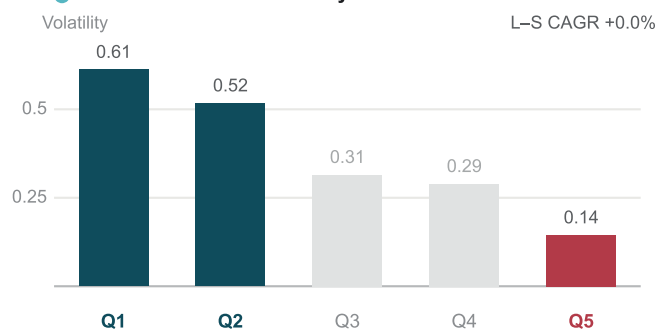
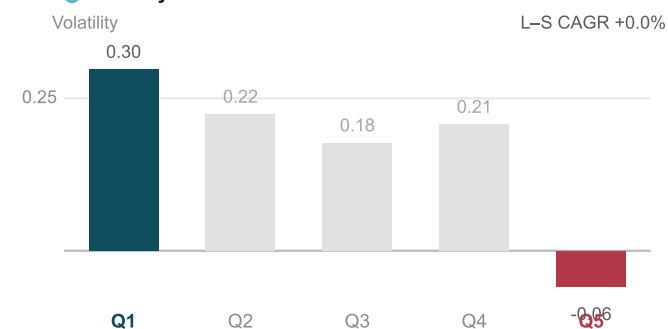
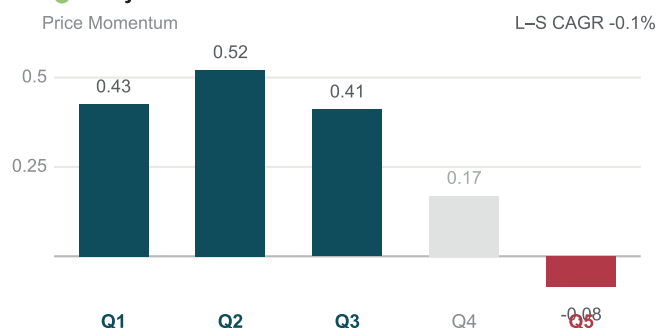
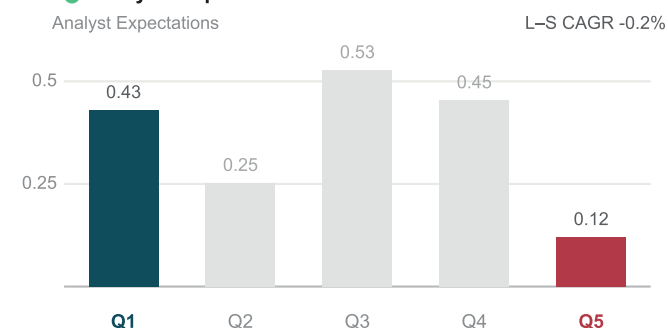
180 ● 1Y Chg in Free Cash Flow to Price



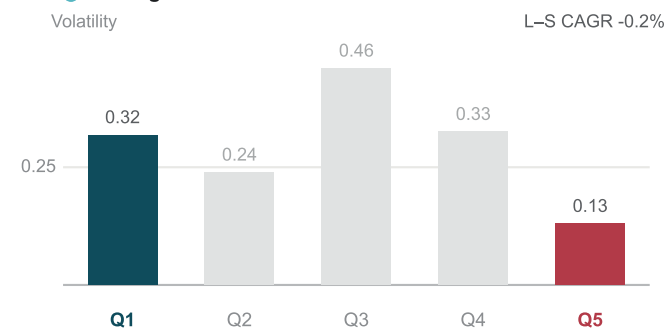
181 ● **39W Lagged Return**182 ● **52W High Low**183 ● **1Y Chg in Shares Outstanding**184 ● **Expected Earnings Growth: Fiscal Year 2/Fiscal Year 1**185 ● **Cash Conversion Cycle**186 ● **Volatility adjusted 12M return**

187 ● 1Y Chg in Sales to Earnings**188 ● Sustainable Growth Rate****189 ● Year over Year Change of Total Debt****190 ● Normalized ROE****191 ● Capital Expenditure to Total Assets****192 ● Trailing 12-Month Overhead/Trailing 12-Month Sales**

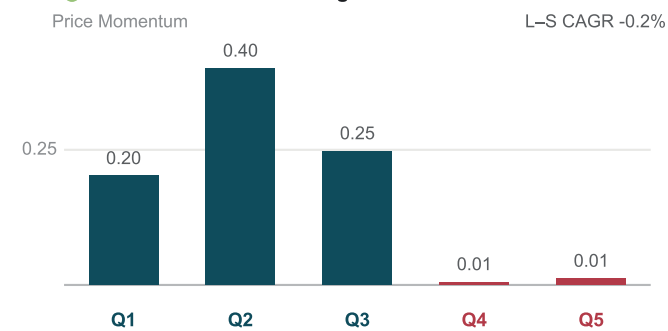


199 ● **Balance Sheet Accruals**200 ● **6M Chg in 12M CAPM Alpha**201 ● **1M Realized Price Volatility**202 ● **130 Day Minimum Return**203 ● **5 Day Price Reversal**204 ● **Analyst Dispersion for FY2 EPS**

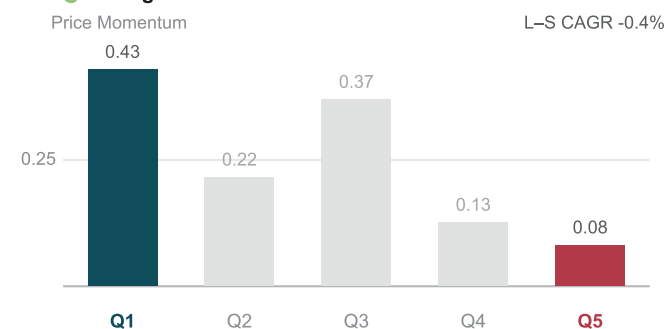
205 ● 1Y Chg in Debt to Assets



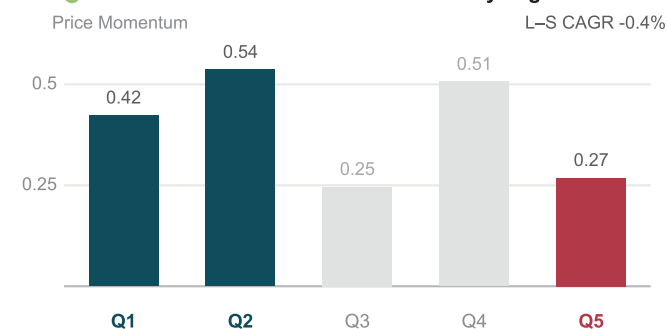
206 ● 26W Relative Price Strength



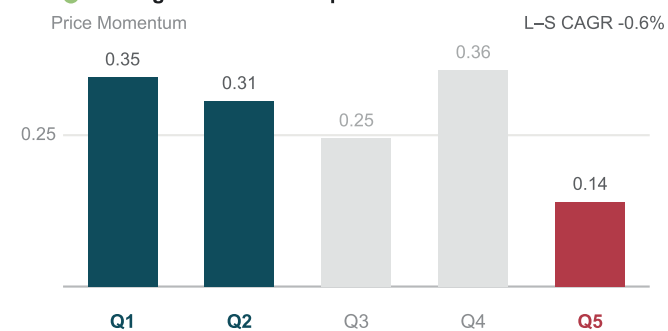
207 ● Change in Number of Holders



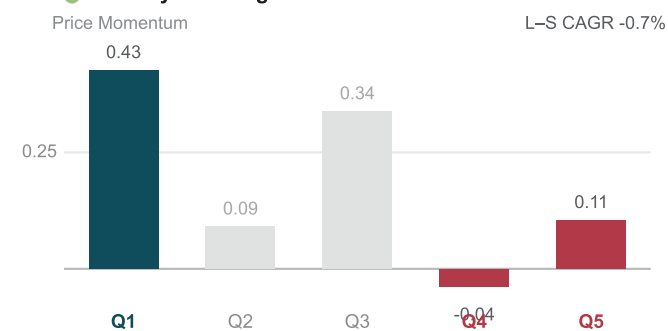
208 ● 52-Week Volume Price Trend with 20-Day Lag

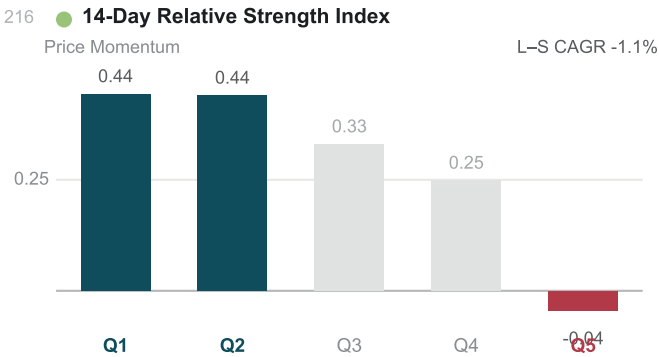
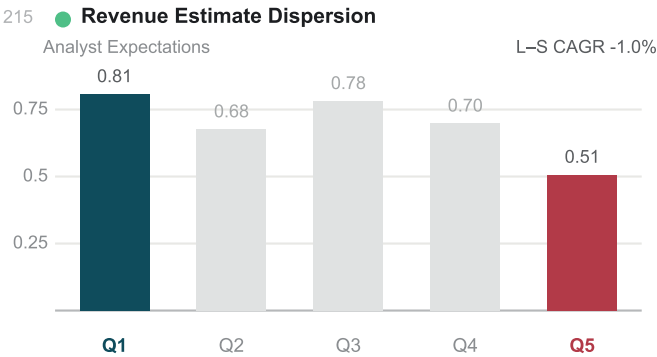
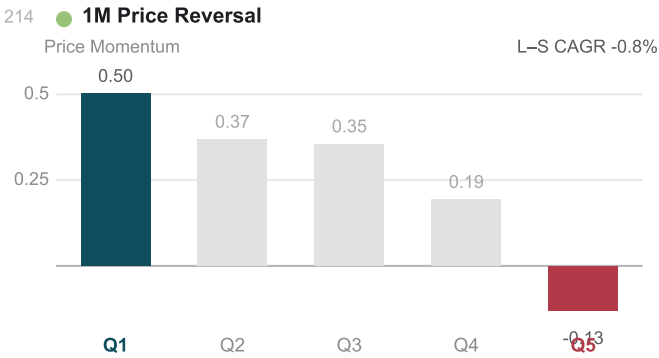
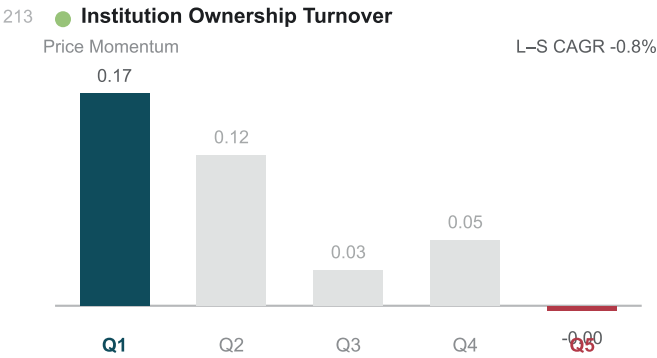
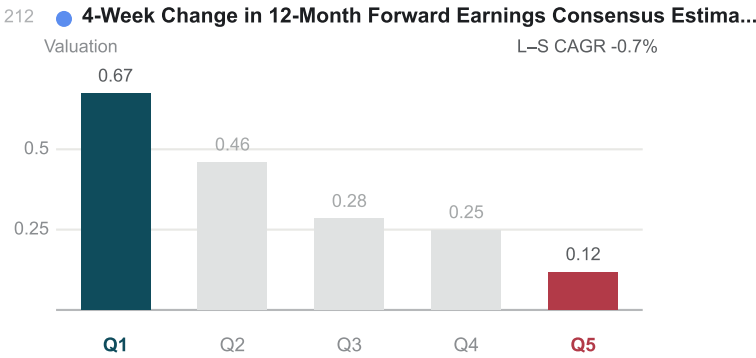
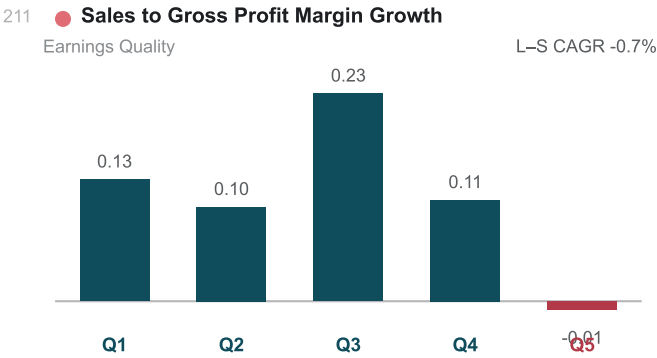


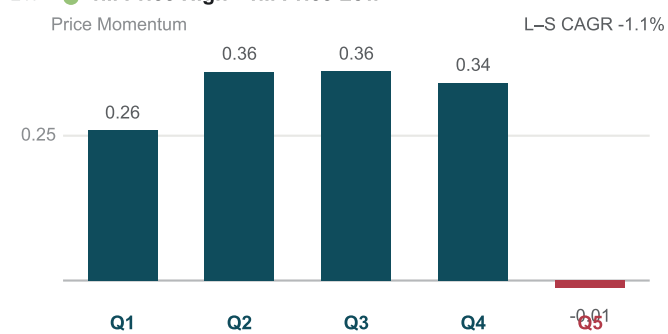
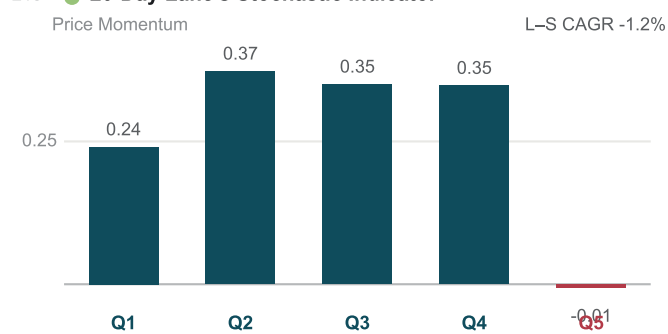
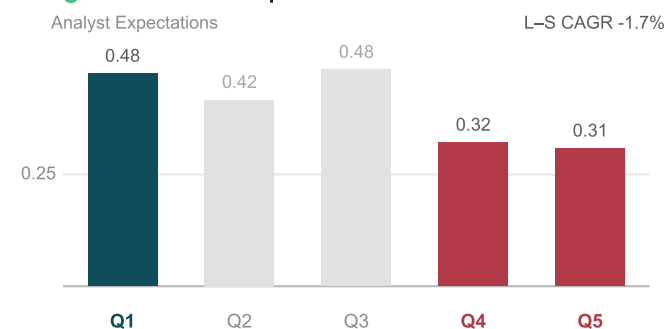
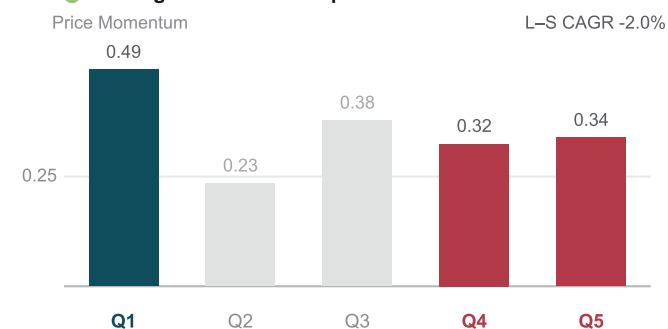
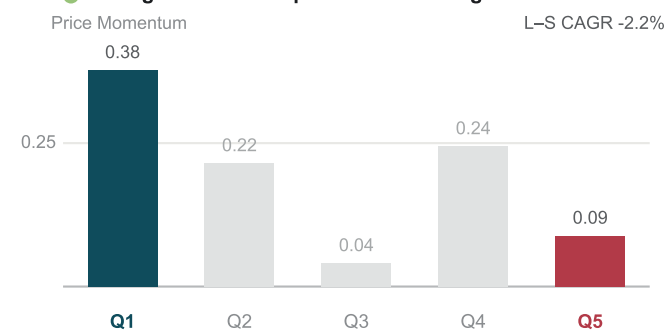
209 ● 6M Chg in 36M CAPM Alpha



210 ● Stability of Change in Number of Holders





217 ● **1M Price High - 1M Price Low**218 ● **20-Day Lane's Stochastic Indicator**219 ● **EBIT Estimate Dispersion**220 ● **6M Chg in 18M CAPM Alpha**221 ● **Change of Ownership Level - All Managers**222 ● **5-Day Money Flow/Volume**